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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHE 411 — Past Questions & Materials
by Renance

Course: CHE 411
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 400

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Paper 1 of 2 — CHE 411 412 413 415 416 AND 417

Level 400 • First Semester • PastQuestion

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ON

Past Questions and Answers

**CHE 411, 412, 413,
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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

FIRST SEMESTER: 2020/2021 SESSION EXAMINATION

COURSE TITLE: SEPARATION PROCESSES II

Credit units: 3

COURSE CODE: CHE 411

Time Allowed: 3 Hours.

INSTRUCTIONS: Answer any five (5) Questions. All Questions carry equal marks

- 1) A mixture of hexane, heptane, and octane is to be separated to give the following products.

Component	Feed		Product		Bottoms		Relative volatility
	F (kmol)	x_F	D (kmol)	x_D	W (kmol)	x_W	
Hexane	40	0.40	40	0.534	0	0	2.7
Heptane	35	0.35	34	0.453	1	0.04	2.22
Octane	25	0.25	1	0.013	24	0.96	1.0

- a) What will be the value of the minimum reflux ratio, if the feed is liquid at its boiling point?
- b) Determine the number of theoretical plates required
- c) Using the attached chart (Figure 1), investigate the change in the number of plates with regards to the reflux ratio.
- 2) A mixture of ortho, meta, and para-methylnitrotoluenes containing 60, 4, and 36 mole per cent respectively of the three isomers is to be continuously distilled to give a top product of 98 mol per cent ortho, and the bottom is to contain 12.5 mole per cent ortho. The mixture is to be distilled at a temperature of 410 K requiring a pressure in the boiler of about 6.0 kN/m². If a reflux ratio of 5 is used, how many ideal plates will be required and what will be the approximate compositions of the product streams? The volatility of ortho relative to the para isomer may be taken as 1.70 and of the meta as 1.16 over the temperature range of 380–415 K
- 3) 160 cm³/s of a solvent S is used to treat 400 cm³/s of a 10 per cent by mass solution of A in B, in a three-stage countercurrent multiple-contact liquid-liquid extraction plant.
- a) What is the composition of the final raffinate?

1

L = 2

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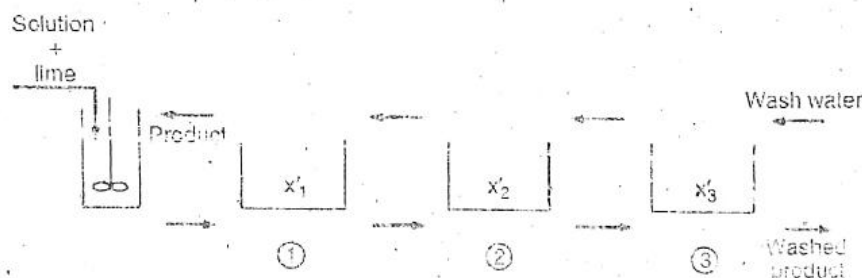
b) Using the same total amount of solvent, evenly distributed between the three stages, what would be the composition of the final raffinate if the equipment were used in a simple multiple-contact arrangement?

The Equilibrium data are as follows:

kg A/kg B:	0.05	0.10	0.15
kg A/kg S:	0.069	0.159	0.258
Densities (kg/m ³):	$\rho_A = 1200$,	$\rho_B = 1000$,	$\rho_S = 80$

4) a) In a pilot scale test using a vessel 1 m³ in volume, a solute was leached from an inert solid and the water was 75 per cent saturated in 100 s. If, in a full-scale unit, 500 kg of the inert solid containing, as before, 28 per cent by mass of the water-soluble component, is agitated with 100 m³ of water, how long will it take for all the solute to dissolve, assuming conditions are equivalent to those in the pilot scale vessel? Water is saturated with the solute at a concentration of 2.5 kg/m³.

b) Caustic soda is manufactured by the lime-soda process. A solution of sodium carbonate in water, containing 0.25 kg/s Na₂CO₃, is treated with the theoretical requirement of lime, and after the reaction is complete the CaCO₃ sludge, containing 1 part of CaCO₃ per 9 parts of water, by mass, is fed continuously to three thickeners in series and washed countercurrently, as shown in the following Figure.



Calculate the necessary rate of feed of neutral water to the thickeners so that the calcium carbonate, on drying, contains only 1 per cent of sodium hydroxide. The solid discharged from each thickener contains 1 part by mass of calcium carbonate to 5 of water. The concentrated wash liquid is mixed with the contents of the agitator before being fed to the first thickener.

5) An ultrafiltration plant is required to treat 50 m³/day of a protein-containing waste stream. The waste contains 0.5 kg/m³ of protein which has to be concentrated to 20 kg/m³ so as to allow recycling to the main process stream. The tubular membranes to be used are available as 30 m² modules. Pilot plant studies show that the flux J through these membranes is given by:

$$J = 0.02 \ln(30/C_f) \text{ m/h.}$$

where C_f is the concentration of protein in kg/m³. Due to fouling, the flux never exceeds 0.04 m/h. Estimate the minimum number of membrane modules required for the operation of this process as.

- a) a single feed and bleed stage.
 b) two feed and bleed stages in series.

Operation for 20 hr/day may be assumed.

6) You have been requested to redesign a water-cooling tower that has a blower with a capacity of $8.30 \times 10^6 \text{ ft}^3/\text{hr}$ of moist air (at 80°F and a wet-bulb temperature of 65°F). The exit air leaves at 95°F and 90°F wet bulb. How much water can be cooled in pounds per hour if the water to be cooled is not recycled, enters the tower at 120°F , and leaves the tower at 90°F ?

7) A single pellet of alumina is exposed to a flow of humid air at a constant temperature. The increase in mass of the pellet is followed automatically, yielding the following results:

t (min)	2	4	10	20	40	60	120
x_r (kg/kg)	0.091	0.097	0.105	0.113	0.125	0.128	0.132

Assuming the effect of the external film is negligible, predict time t against x_r values for a pellet of twice the radius, where x_r is the mass of adsorbed phase per unit mass of adsorbent.

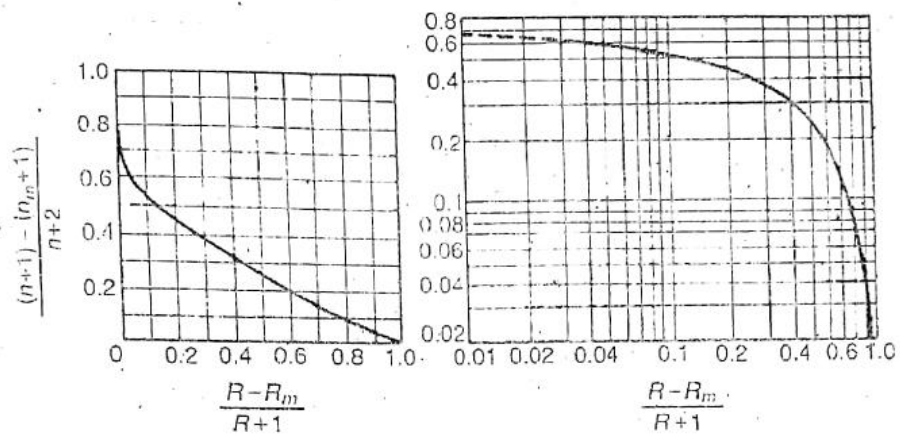
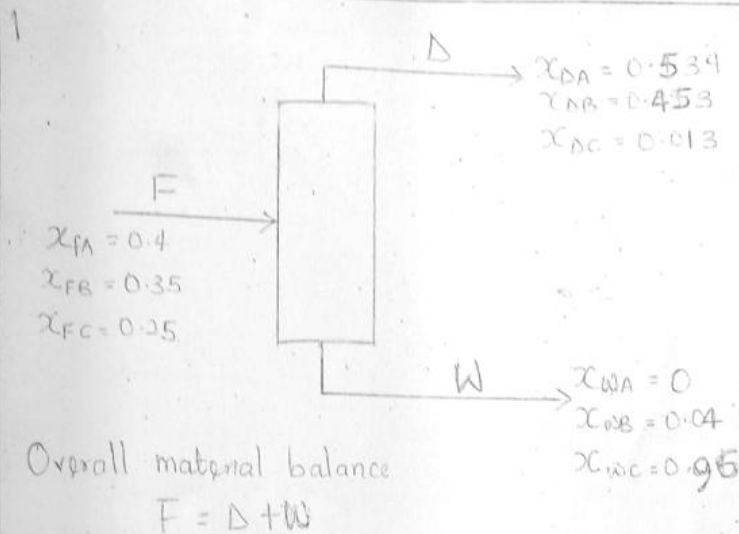


Figure 1: Relationship between Reflux ratio and Number of Plates

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CHE 411 (2020/2021)



a. Minimum reflux ratio when feed is liquid, R_m

Using the formula: $\frac{\alpha_A x_{FA}}{\alpha_A - \theta} + \frac{\alpha_B x_{FB}}{\alpha_B - \theta} + \frac{\alpha_C x_{FC}}{\alpha_C - \theta} + \dots = 1 - q$

where $\alpha_A = 2.70$, $\alpha_B = 2.22$, $\alpha_C = 1.0$, $q = 1$ (feed is liquid)

$$\left(\frac{2.70 \times 0.4}{2.70 - \theta} \right) + \left(\frac{2.22 \times 0.35}{2.22 - \theta} \right) + \left(\frac{1 \times 0.25}{1 - \theta} \right) = 1 - 1 = 0$$

The required value of θ must satisfy the relation $\alpha_B < \theta < \alpha_A$

That is $1.0 < \theta < 2.22$ where light key (A) is heptane and heavy key (B) is octane.

Assuming $\theta = 1.17$ (after trying values of θ between this range)

$$\text{then } \left(\frac{2.70 \times 0.4}{2.7 - 1.17} \right) + \left(\frac{2.22 \times 0.35}{2.22 - 1.17} \right) + \left(\frac{1 \times 0.25}{1 - 1.17} \right) = 0$$

To get the minimum ratio, R_m

$$\frac{\alpha_A x_{DA}}{\alpha_A - \theta} + \frac{\alpha_B x_{DB}}{\alpha_B - \theta} + \frac{\alpha_C x_{DC}}{\alpha_C - \theta} + \dots = R_m + 1$$

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$$\left(\frac{2.22 \times 0.453}{2.22 - 1.17} \right) + \left(\frac{1.00 \times 0.013}{1.00 - 1.17} \right) = 1.827$$

$$1.827 = R_m + 1$$

$$R_m = 1.827 - 1 = \underline{0.827}$$

b. Number of theoretical plates required

This can be found by applying Fenske's equation to the key components (Heptane and Octane)

$$n+1 = \frac{\log \left[\left(\frac{x_A}{x_B} \right)_d \left(\frac{x_B}{x_A} \right)_w \right]}{\log(\alpha_{AB})}$$

$$\alpha_{AB} = \frac{\alpha_A}{\alpha_B} = \frac{2.22}{1} = 2.22$$

n is the number of plates

$$n+1 = \frac{-\log \left[\frac{0.0453}{0.013} \times \frac{0.96}{0.04} \right]}{-\log 2.22} = \frac{\log 83.5}{\log 2.22} = 8.5$$

$$n+1 = 8.5 ; n = 8.5 - 1 = 7.5 \approx 8$$

Number of theoretical plates = 8

c. To investigate the change in number of plates, n with the reflux ratio, R

$$\text{Since } R_m = 0.8 \text{ and } n+1 = 8.5$$

$$\text{Using the chart in Fig 1 } \left(\frac{(n+1) - (n_{m+1})}{n+2} \right) \text{ vs } \frac{R - R_m}{R+1}$$

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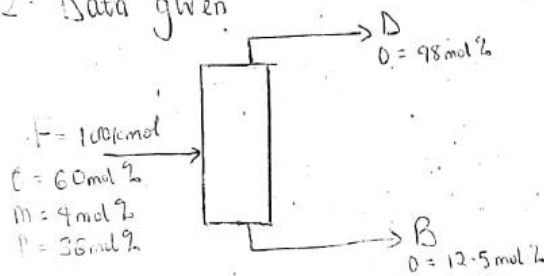
R	$\frac{R - R_m}{R + 1}$	$\frac{(n+1) - (n_m+1)}{n+2}$	n
1	$\frac{1 - 0.83}{1 + 1} = 0.085$	0.55	19
2	$\frac{2 - 0.83}{2 + 1} = 0.390$	0.32	11.8
5	$\frac{5 - 0.83}{5 + 1} = 0.695$	0.15	9.1
10	$\frac{10 - 0.83}{10 + 1} = 0.833$	0.08	8.2

NB: The values from $(n+1) - (n_m+1)$ is gotten from the graph in Fig 10.

Thus making n the subject of formula, we get the values of n .

The number of plates falls off rapidly at first, though more slowly later, and a value of $R \approx 2$ is probably the most economic.

2. Data given



Total material Balance

$$F = D + B$$

$$100 = D + B$$

Component balance on ortho

$$X_{F0}F = X_{D0}D + X_{B0}B$$

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$$100 \times 0.6 = 0.98D + 0.125B$$

$$60 = 0.98D + 0.125B$$

$$\text{Recall: } 100 = D + B; \quad D = 100 - B$$

$$60 = 0.98(100 - B) + 0.125B$$

$$B = 44.44 \text{ kmol}$$

$$D = 100 - B = 100 - 44.44$$

$$D = 55.56 \text{ kmol}$$

The compositions and amounts of the streams are:

Component	Feed		Distillate		Bottoms	
	kmol	mol%	kmol	mol%	kmol	mol%
ortho	60	60	54.44	98.0	5.56	12.5
meta	4	4	0.33	0.6	3.67	8.3
para	36	36	0.79	1.4	35.21	79.2
	100	100	55.56	100	44.44	100

Equations of operating lines

The liquid and vapour streams in the columns are obtained as follows:

$$\text{Above the feed point: Liquid downflow } L_n = 5D = 277.8$$

$$\text{Vapour up, } V_n = 6D = 333.4$$

Below the feed point, assuming the feed is liquid at its boiling point:

$$\text{Liquid downflow, } L_m = L_n + F = 277.8 + 100 = 377.8$$

$$\text{Vapour up, } V_m = L_m - W = 377.8 - 44.44 = 333.4$$

The equations for the operating lines may then be written as:

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Below the feed plate:

$$y_m = \frac{L_m}{V_m} x_{m+1} - \frac{W}{V_m} x_w$$

ortho: $y_{mo} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.125)$

$$y_{mo} = 1.133 x_{m+1} - 0.0166$$

meta: $y_{mm} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.083)$

$$y_{mm} = 1.133 x_{m+1} - 0.011$$

Para: $y_{mp} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.792)$

$$y_{mp} = 1.133 x_{m+1} - 0.105$$

Above the feed plate:

$$y_n = \frac{L_n}{V_n} x_{n+1} + \frac{D}{V_n} x_d$$

ortho: $y_{no} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.98)$

$$y_{no} = 0.833 x_{n+1} + 0.163$$

meta: $y_{nm} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.006)$

$$y_{nm} = 0.833 x_{n+1} + 0.001$$

Para: $y_{np} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.014)$

$$y_{np} = 0.833 x_{n+1} + 0.002$$

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Composition of liquid on first plate

$$y_{s1} = x_{s1} / \sum x_{s1}$$

$$0.191 = 1.133x_1 + 0.0166$$

$$x_1 = 0.183$$

Using iteration method, we have 16 ideal plates

3) Equilibrium data:

kg A / kg B	0.05	0.10	0.15
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kg A / kg B	0.069	0.159	0.258
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Densities (kg/m ³)	$\rho_A = 1200$	$\rho_B = 1000$	$\rho_S = 800$
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a) Countercurrent operation

Considering the solvent S, $160 \text{ cm}^3/\text{s} = 1.6 \times 10^{-4} \text{ m}^3/\text{s}$

and mass flowrate = $(1.6 \times 10^{-4} \times 800) = 0.128 \text{ kg/s}$

Solution, $400 \text{ cm}^3/\text{s} = 4 \times 10^{-4} \text{ m}^3/\text{s}$

containing $a \text{ m}^3/\text{s}$ A and $(5 \times 10^{-4} - a) \text{ m}^3/\text{s}$ B

Thus: mass flowrate of A = $1200a \text{ kg/s}$

mass flowrate of B = $(4 \times 10^{-4} - a)1000 = (0.4 - 1000a) \text{ kg/s}$

Total = $(0.4 + 2000a) \text{ kg/s}$

The concentration of the solution is

$$0.10 = \frac{1200a}{0.4 + 2000a}$$

$$\therefore a = 3.39 \times 10^{-5} \text{ m}^3/\text{s}$$

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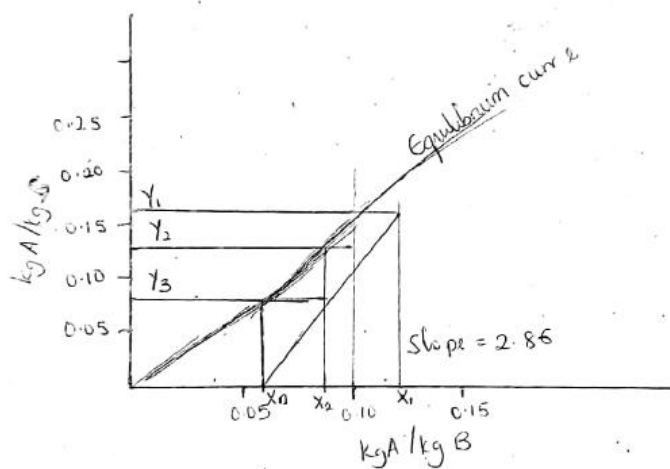
$$\therefore \text{mass flowrate of A} = 1200 \times 3.39 \times 10^{-5} \\ = 0.041 \text{ kg/s}$$

$$\text{mass flowrate of B} = (0.4 - 1000 \times 3.39 \times 10^{-5}) \\ = 0.366 \text{ kg/s}$$

$$\text{Ratio of A to B in the feed} = \frac{0.041}{0.366} = 0.112 \text{ kg/kg}$$

The ratio, $x_f = 0.112 \text{ kg/kg}$ is marked in.

$$\frac{\text{mass flowrate of B}}{\text{mass flowrate of S}} = \frac{0.366}{0.128} = 2.86$$



Since pure solvent is added $y_{n+1} = y_n = 0$ and a line of slope 2.86 is drawn in such that off from $x_f = 0.112 \text{ kg/kg}$ to $y_4 = 0$ gives exactly three stages.

$$\text{When } y_4 = 0, x_n = x_3 = 0.057 \text{ kg/kg}$$

Hence, the composition of final raffinate = 0.057 kg A/kg B

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∴ multiple contact

∴ $\frac{0.128}{3} = 0.0427 \text{ kg/s}$ of pure solvent S is fed to each stage

Stage 1 $x_f = \frac{0.041}{0.366} = 0.112 \text{ kg/kg}$

From the equilibrium curve, the extract contains 0.18 A/kg S and
 $0.18 \times 0.0427 = 0.0077 \text{ kg/s A}$

Raffinate from stage 1 contains: $0.041 - 0.0077 = 0.0333 \text{ kg/s A}$
 0.366 kg/s B

$$x_i = \frac{(0.0333)}{0.366} = 0.091 \text{ kg/kg}$$

Stage 2

$$x_i = 0.091 \text{ kg/kg}$$

From graph, this contains 0.14 kg A/kg S
 $\therefore 0.14 \times 0.0427 = 0.006 \text{ kg/s A}$

The raffinate from stage 2 contains: $0.0333 - 0.006 = 0.0273 \text{ kg}$
and 0.366 kg/s B

$$x_2 = \frac{0.0273}{0.366} = 0.075 \text{ kg/kg}$$

Stage 3

$$x_2 = 0.075 \text{ kg/kg}$$

The extract contains 0.114 kg A/kg S
 $0.114 \times 0.0427 = 0.0049 \text{ kg/s A}$

The raffinate from stage 3 contains: $0.0273 - 0.0049 = 0.0224 \text{ kg/}$
and 0.366 kg/s B

$$x_3 = \frac{0.0224}{0.366} = 0.061 \text{ kg/kg (final composition)}$$

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4(a) For the pilot scale vessel:

$$C = \left(2.5 \times \frac{28}{100} \right) = 1.875 \text{ kg/m}^3$$

$$C_s = 2.5 \text{ kg/m}^3 \quad V = 1.0 \text{ m}^3 \quad t = 10 \text{ s}$$

$$C = C_s (1 - e^{-(k'A/v)t})$$

$$1.875 = 2.5 (1 - e^{-(\frac{k'A}{100})100})$$

$$\frac{k'A}{b} = 0.139 \text{ m}^2/\text{s}$$

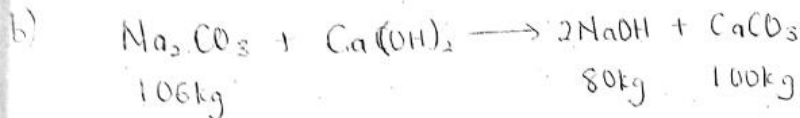
For the full scale vessel

$$C = \left(2.5 \times \frac{28}{100} \right) = 1.40 \text{ kg/m}^3$$

$$C_s = 2.5 \text{ kg/m}^3 \quad V = 100 \text{ m}^3$$

$$1.40 = 2.5 (1 - e^{-(\frac{0.139t}{100})})$$

$$t = 591 \text{ s } (9.9 \text{ mins})$$



If x'_1 , x'_2 and x'_3 are the solute: solvent ratios in thickness 1, 2, and 3 respectively, the quantities of CaCO_3 , NaOH and water in each

of the streams can be calculated for every 100 kg of calcium Carbonate

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	Calcs	NaOH	Water
Overall balance			
Feed from reactor	100	80	900
Feed as wash water	—	—	w_f
Product underflow	100	$300x'_3$	300
Product overflow	—	$80 - 300x'_3$	$600 + w_f$
Thickener 1			
Feed from reactor	100	80	900
Feed overflow	—	$300(x'_1 - x'_3)$	w_f
Product underflow	100	$300x'_1$	300
Product overflow	—	$80 - 300x'_3$	$600 + w_f$
Thickener 2			
Feed underflow	100	$300x'_1$	300
Feed overflow	—	$300(x'_2 - x'_3)$	w_f
Product underflow	100	$300x'_2$	300
Product overflow	—	$300(x'_1 - x'_3)$	w_f
Thickener 3			
Feed underflow	100	$300x'_2$	300
Feed water	—	—	w_f
Product underflow	100	$300x'_3$	300
Product overflow	—	$300(x'_2 - x'_3)$	w_f

Since the final underflow must contain only 1 percent of NaOH

$$\text{then: } \frac{300x'_3}{100} = 0.01$$

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If the equilibrium is achieved in each of the thickeners, the ratio of NaOH to water will be the same in the underflow and the overflow and:

$$\frac{300(x'_2 - x'_3)}{W_f} = x'_3$$

$$\frac{300(x'_1 - x'_3)}{W_f} = x'_2$$

$$\frac{80 - 300x'_3}{600 + W_f} = x'_1$$

Solution of these four simultaneous equations gives:

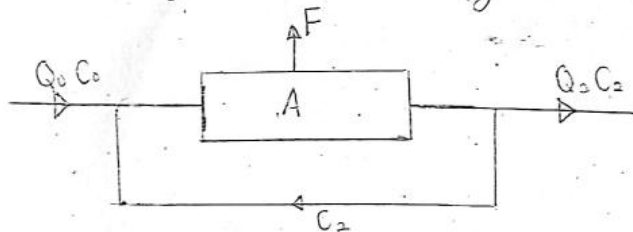
$$x'_3 = 0.0033 \quad x'_2 = 0.0142 \quad x'_1 = 0.05 \quad W_f = 980$$

The amount of water required for washing 100 kg CaCO_3 is 980 kg

The solution fed to reactor contains 0.25 kg Na_2CO_3 . This is equivalent to 0.236 kg/l CaCO_3 and hence the actual water required

$$= \frac{980 \times 0.236}{100} = 0.23 \text{ kg/l}$$

5 (a) With a single feed and bleed stage



Where Q_0 = Volumetric flowrate of feed

Q_2 = Volumetric flowrate of concentrate

C_0 = Solute concentration in the feed

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A = required membrane area

NB: It is assumed that there is no loss of solute through the membrane.

The concentration (C_1) at which the flux becomes fouling-limited is:

$$0.04 = 0.02 \ln \left(\frac{30}{C_1} \right)$$

$$C_1 = 4 \text{ kg/m}^3$$

That is, below this concentration the membrane flux is 0.04 m/h .

This does not pose a constraint for the single stage as the concentration of solute C_2 will be that of the final concentration, 20 kg/m^3 .

Conservation of solute gives: $Q_0 C_0 = Q_2 C_2$

Fluid balance: $Q_0 = F + Q_2$

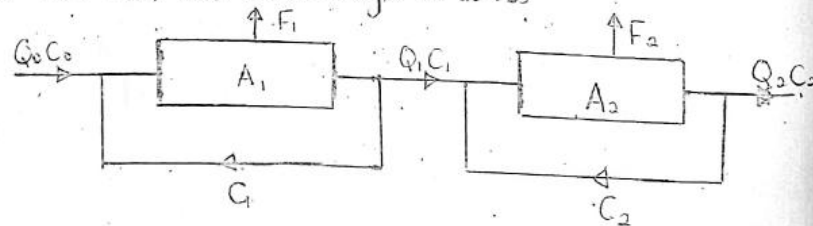
Combining the equations and substituting known values

$$2.438 = A \cdot 0.02 \ln \left(\frac{30}{20} \right)$$

$$A = 30 \text{ m}^2$$

Thus 10 modules will almost meet the specification for the single stage.

b) With two feed and bleed stages in series



Q_1 = Volumetric flowrate of retentate at the intermediate point

C_1 = Concentration of solute in the retentate.

A_1 and A_2 are the area of membrane.

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Conservation of solute: $Q_0 C_0 = Q_1 C_1 = Q_2 C_2$

A fluid balance on stage 1: $Q_0 = Q_1 + F_1$

A fluid balance on stage 2: $Q_1 = Q_2 + F_2$

Substituting given values in equations above

$$2.5 = \frac{1.25}{C_1} + 0.02 A_1 \ln\left(\frac{30}{C_1}\right)$$

$$\frac{1.25}{C_1} = 0.0625 + 0.00811 A_1$$

The procedure is to use trial by error to eliminate the value of C_1 that gives the optimum values of A_1 and A_2 . Thus,

If $C_1 = 5 \text{ kg/m}^3$, then $A_1 = 63 \text{ m}^2$ and $A_2 = 23 \text{ m}^2$

An arrangement of 3 modules - 1 module is required

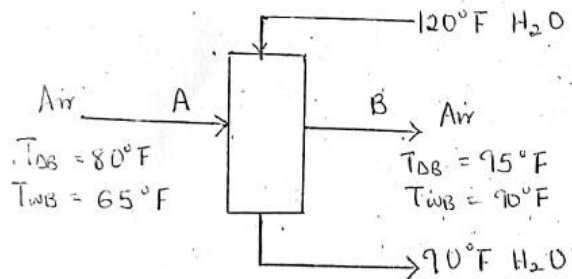
If $C_1 = 4 \text{ kg/m}^3$, then $A_1 = 55 \text{ m}^2$ and $A_2 = 31 \text{ m}^2$

An arrangement of 2 modules - 1 module is almost sufficient

If $C_1 = 4.5 \text{ kg/m}^3$, then $A_1 = 59 \text{ m}^2$ and $A_2 = 27 \text{ m}^2$

An arrangement of 2 modules - 1 module which meets the requirement

6)



A = Inlet air, B = Exit air

T_{db} = Dry bulb temperature

T_{wb} = Wet bulb temperature

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From the psychrometric chart, we can determine the amount of water in dry air, enthalpy, humidity and humid volume for both A and B, which are given as:

	A	B
Water in dry air = $\frac{\text{ft}^3 \left(\frac{\text{lb H}_2\text{O}}{\text{lb dry air}} \right)}$	0.0098	0.0297
Enthalpy = $\Delta H \left(\frac{\text{Btu}}{\text{lb dry air}} \right)$	29.93	55.83
Humidity = $\left(\frac{\text{g moist H}_2\text{O}}{\text{lb dry air}} \right)$	69	208
Humid volume = $\hat{V} \left(\frac{\text{ft}^3}{\text{lb dry air}} \right)$	13.82	14.65

In calculating for the cooling water exit rate, we have to determine an energy balance around the process:

Given the basis: $8.30 \times 10^6 \text{ ft}^3/\text{hr}$ of moist air

Converting the basis to lb/hr

$$\frac{8.30 \times 10^6 \text{ ft}^3}{\text{hr}} \times \frac{\text{lb dry air}}{13.82 \text{ ft}^3} = 6.00 \times 10^5 \text{ lb dry air/hr}$$

Finding the relative enthalpy of the entering water stream, taking the reference temperature of water to be 32°F (0°C), using the enthalpy equation which is:

$$\Delta H = C_{p_{H_2O}} \cdot \Delta T, \quad C_{p_{H_2O}} = 1$$

$$\Delta H = 1(120 - 32) = 88 \text{ Btu/lb H}_2\text{O}$$

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The relative enthalpy of the exit stream

$$\Delta H = C_{p_{H_2O}} \Delta T = 1(90-32) = 58 \text{ Btu/lb } H_2O.$$

Taking the loss of water through the cooling tower to be

$$0.0297 - 0.0098 = 0.0199 \text{ lb } H_2O/\text{lb dry air}$$

Taking the material balance of the water stream to be

$$1 - 0.0199 = \frac{\text{lb } H_2O \text{ leaving tower in the water stream}}{\text{lb dry air}}$$

Taking the energy/enthalpy balance around the entire process to be

air and water in air entering + water stream entering =

air and water in air leaving + water stream leaving.

$$\begin{array}{c} 29.93 \text{ Btu} \\ \text{lb dry air} \end{array} \left| \frac{6.00 \times 10^3 \text{ lb dry air}}{\text{lb dry air}} \right. + \begin{array}{c} 88 \text{ Btu} \\ \text{lb } H_2O \end{array} \left| \frac{w \text{ lb } H_2O}{\text{lb dry air}} \right| \frac{6 \times 10^3 \text{ lb dry air}}{\text{lb dry air}}$$

$$= \begin{array}{c} 55.83 \text{ Btu} \\ \text{lb dry air} \end{array} \left| \frac{6 \times 10^3 \text{ lb dry air}}{\text{lb dry air}} \right. + \begin{array}{c} 58 \text{ Btu} \\ \text{lb } H_2O \end{array} \left| \frac{(w - 0.0199) \text{ lb } H_2O}{\text{lb dry air}} \right| \frac{6 \times 10^3 \text{ lb dry air}}{\text{lb dry air}}$$

$$(29.93 + 88w)(\text{Btu})(6 \times 10^3) = 55.83 + 58(w - 0.0199)(6 \times 10^3)(\text{Btu})$$

$$29.93 + 88w = 55.83 + 58(w - 0.0199)$$

$$w = 0.825 \text{ lb } H_2O/\text{lb dry air}$$

There the amount of water cooled in lb/hr will be given as:

$$\begin{array}{c} 0.825 \text{ lb } H_2O \\ \text{lb dry air} \end{array} \left| \frac{6 \times 10^3 \text{ lb dry air}}{\text{hr}} \right. = 4.83 \times 10^5 \text{ lb } H_2O/\text{hr}$$

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1) When pellet diffusion is dominant, assuming $C_{so} = 0$ and the concentration outside the pellet is constant, then

$$\frac{C_s}{C_s^*} = 1 - \frac{6}{\pi^2} \exp \left[-\left(D_R \pi^2 t / r_i^2 \right) \right]$$

Hence a plot of $\ln \left[1 - \frac{C_s}{C_s^*} \right]$ against t should be linear.

For $r = r_i$ with $C_s^* = 0.132 \text{ kg/kg}$ then

$t \text{ (min)}$	$C_s \text{ (kg/kg)}$	C_s/C_s^*	$1 - (C_s/C_s^*)$
2	0.091	0.69	0.31
4	0.097	0.73	0.27
10	0.105	0.80	0.20
20	0.113	0.86	0.14
40	0.125	0.95	0.05
60	0.128	0.97	0.03
120	0.132	1.0	0

The data are plotted in a graph which confirms the linearity and from which $\frac{\pi^2 D_R}{r_i^2} = 0.043$

For a pellet of twice the radius, that is $r = 2r_i$

$$\text{and the slope} = (-0.043/4) = -0.011$$

Thus, when the radius = $2r_i$

$$\frac{C_s}{C_s^*} = 1 - \frac{6}{\pi^2} \exp(0.011t)$$

Alternatively, use may be made of the quadratic driving force equation

$$\frac{dC_s}{dt} = \frac{k D_R}{r_i^2} \cdot \frac{C_s^{*2} - C_s^2}{2(C_s - C_{so})} \quad \text{--- (1)}$$

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Integrating from the initial condition, $t=0$ and $C_s=0$, then:

$$\frac{C_s}{C_s^*} = \left[1 - \exp\left(-\frac{k D_e t}{r_p^2}\right) \right]^{0.5}$$

Indicating that a plot of $\ln \left[1 - \left(\frac{C_s}{C_s^*} \right)^2 \right]$ against t should also be linear

t (min)	C_s (kg/kg)	(C_s/C_s^*)	$1 - \left[\frac{C_s}{C_s^*} \right]^2$
2	0.091	0.69	0.52
4	0.097	0.73	0.47
10	0.105	0.80	0.36
20	0.113	0.86	0.26
40	0.125	0.95	0.10
60	0.128	0.97	0.06
120	0.132	1.0	0

Therefore: $\frac{k D_e}{r_p^2} = 0.04$

For a pellet twice the size $= \frac{C_s}{C_s^*} = \left[1 - \exp(-0.01t) \right]^{0.5}$ — (i)

Values of C_s for radius $= 2r_p$ are calculated from equ. (i) and (ii) to give the following results

t (mins)	C_s (kg/kg)	
	equ (i)	equ (ii)
4	0.055	0.026
20	0.068	0.056
60	0.091	0.088

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING
First Semester Examination 2020/2021 Session

COURSE TITLE: Process Instrumentation
COURSE CODE: CHE 412

Credit Unit: 2
Time Allowed: 2 hrs

INSTRUCTIONS: Answer *FOUR (4)* questions in all.

- (1) (a) Clarify what a Process Industry is and state five examples.
(b) State one raw material and one finished product for each of the five examples of process industry in (1a) above.
(c) Explain what determines whether the processing of a raw material will be simple, complex, and potentially hazardous?
- (2) (a) Define (i) Process Control and (ii) Process Variable.
(b) (i) List five examples of process variables (ii) Explain the three basic reasons why a process is controlled.
(c) Describe the following systems and state one example where each is applied: (i) Sequential control (ii) Elementary on/off action and (iii) Continuous control action.
- (3) (a) Describe the following terms: (i) Measured variable (ii) Manipulated variable and (iii) Error
(b) Define the following terms: (i) Control algorithms (ii) Off set and (iii) Load disturbance
(c) Explain the three major components of error.
- (4) (a) Distinguish between manual and automatic control systems.
(b) (i) What is a primary element in control systems? (ii) Give five examples of primary elements.
(c) (i) Write the following abbreviations in full: DP, RTD, HVAC and LRV (ii) Describe the operation of an RTD.
- (5) (a) Define the following, and give one example of each (i) Transducer (ii) Controller
(b) (i) What is a transmitter? (ii) List five examples of a transmitter.
(c) (i) What is a signal in the process industry? (ii) State and explain the three types of signals in a control system.
- (6) (a) (i) What is protocol? (ii) State and explain the two categories of protocols.
(b) Describe the following, and give one example of each (i) Indicator (ii) Recorder (iii) Controller
(c) (i) What is Final Control element in a control system? (ii) List three examples of final control element. (iii) A pressure gauge located at the base of an open tank containing a liquid with specific weight 13.6 KN/m^3 , recorded 1.27 MPa . What is the depth of the fluid in the tank?

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CHE 412 2020/2021

(1)

a) This is an industry, such as the chemical or petrochemical industry, that is concerned with the processing of bulk resources into other products i.e. industries where the primary production processes are either continuous, or occur on a batch of materials that is indistinguishable.

Examples

- 1) food and beverage industries
- 2) ceramics
- 3) wood and wood products
- 4) chemicals & pharmaceuticals
- 5) petroleum

Industry	Raw material	Finished products	NB: choose any one of the available examples
Food	oil palm	palm oil, mats, baskets	
Ceramics	Silica	Tiles, clay	
Wood and wood products	wood from trees	chairs, tables, papers, construction materials	
Pharmaceuticals	mono chloro benzene (mcb)	Antibiotics, vaccines, human blood-plasma fractions	
Petrochemical	crude oil	Gasoline, jet-fuel, waxes, lubricating oil, asphalt etc.	

The processing of raw materials into desired end products can be simple, complex and potentially hazardous, depending on the raw materials being used and in the manner they are handled. It is therefore important to control a process in order to achieve the desired end products in a safe and efficient manner.

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②

2a) i) PROCESS CONTROL

This refers to the methods that are used in control process variables in a process industry.

ii) PROCESS VARIABLES

These are parameters of a process that can cause change in the process condition when altered.

ii) Examples of process variables

- 1) Pressure
- 2) Temperature
- 3) Level
- 4) pH
- 5) Flow rate

ii) BASIC REASONS WHY A PROCESS IS CONTROLLED

- 1) Reduce variability
- 2) Increased efficiency
- 3) Ensure safety

i) REDUCE VARIABILITY

Process control helps in reducing the variability in various batches of end products and in avoiding padding.

padding is the process of carrying out additional work on finished product in order to meet specifications. accurate and reliable process control, variability in product specification is reduced, and quality ensured, also profit margin increased.

ii) INCREASED EFFICIENCY

For processes that requires being maintained at certain set points or desired optimal points, the use of dependable process controls will ensure process efficiency and increased profit margin.

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Basic Safety

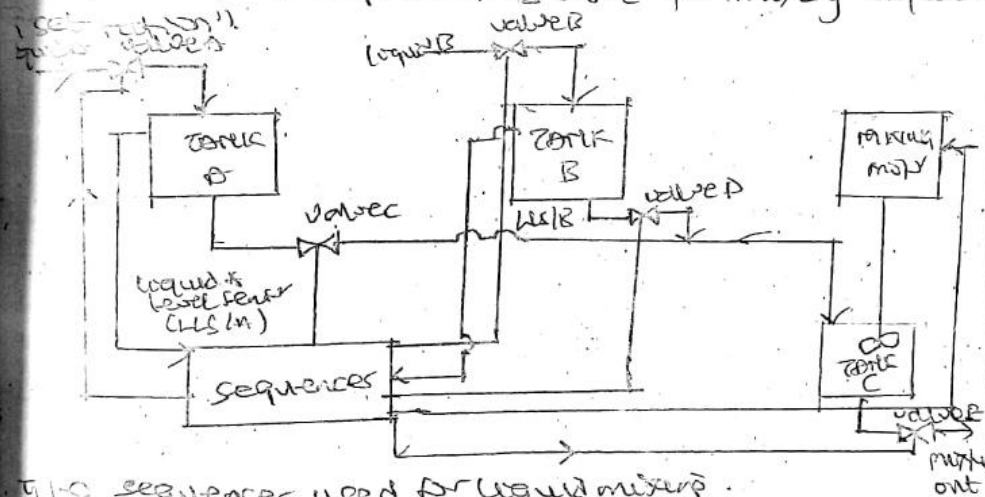
or processes that have no dependable process control may experience catastrophic safety issues, such as un-away reactions, explosions, implosions (an unusual violent collapsing of a vessel), fire, leakage, polluter etc. The financial and reputational losses from safety issues can be very huge to an organization.

1. Sequential Control

Sequential control is an event based process, in which completion of one event follows the completion of another, until a process is complete.

It is achieved by a sensing device which gives feedback signal from each step of the process.

An example of sequential control system is shown above, where a sequencer is used for mixing liquids.



The sequencer used for liquid mixing.

Sequence of Events

- 1. Open valve A to fill Tank A
- 2. When Tank A is full, a feedback signal from the full level sensor [LLS] sends signal to the sequencer to turn off valve A.
- 3. Open valve B to fill Tank B
- 4. When Tank B is full, a feedback signal from the full level sensor sends signal to the sequencer to

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- 5) turn off valve B.
 - 6) When valves A and B are closed, valves C and D are opened, to allow measured quantities of liquid A and liquid B flow in tank C, the mixing tank.
 - 7) After valve C and D are closed, start mixing motor to run for a set period.
 - 8) Turn off mixing motor.
 - 9) Open valve E, to flow out mixture.
 - 10) Turn off valve E, when the signal is received.
- This process can be repeated from step 5 until Tank A and Tank B are empty, before starting all over from step 1.

(i) Elementary ON/OFF ACTION

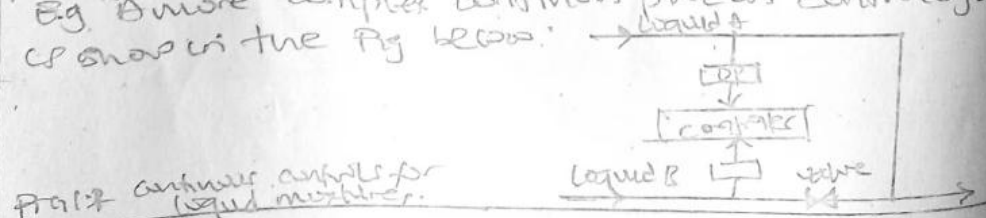
This is a category of "continuous process control", & Elementary on/off action system only has capability to turn on and turn off an action. eg it is mainly used in heating, ventilation and air conditioning (HVAC) systems where the process variable does not require to be measured with great precision. It is suited for intermittent actions.

(ii) Continuous control action

In continuous control actions, it is required to continually control a physical output parameter of a process. The parameter is measured with an instrument or sensor, compared to a set value.

Any deviation between the two values causes an error signal to be generated, which is then used in adjusting an input parameter to correct the output change.

Eg. A more complex continuous process control system is shown in the fig below:



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Fig 1-1, where a mixture of two liquids is required, the rate of liquid A is measured with differential pressure (DP) sensor, and the amplitude of the signal - in the differential pressure is used by the controller - reference signal (or set point) to control the flow rate of liquid B.

The controller uses a DP sensor to measure the flow rate of liquid B, and compares its amplitude to that of the monitoring flow of liquid A.

The difference between the two signals (which is error) is used in controlling the valve, such that the rate of liquid B (manipulating variable) is directly proportional to that of liquid A - thus the two liquids combined in a set ratio.

(3)

1) MEASURED VARIABLE

The measured variable is the electronic or pneumatic representation of the value of the controlled variable. The measured variable typically comes from a transmitter, which measures the controlled variable and produces an output representative of it.

2) MANIPULATED VARIABLE

It is a variable that we can control in the process and which affects the output of the process. It is the one we can change, either increasing it or decreasing it.

3) Error

In the process control system, error is defined as the difference between set point and process variable and given by the equation

$$\text{Error} = \text{set point} - \text{process variable}$$

where the set point is the desired value.

$$\text{Maximum value} = \text{set point} + \text{error}$$

$$\text{Minimum value} = \text{set point} - \text{error}$$

$$\text{Range} = \text{max value} - \text{min value} \therefore \text{Error} = \frac{\text{Range}}{2}$$

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B) CONTROL ALGORITHMS

It is a mathematical expression of a control function eg $U = f(\pm e)$

Using the temperature loop example, U in the above equation is the fuel valve position and e is the error. See Fig 1.2 below

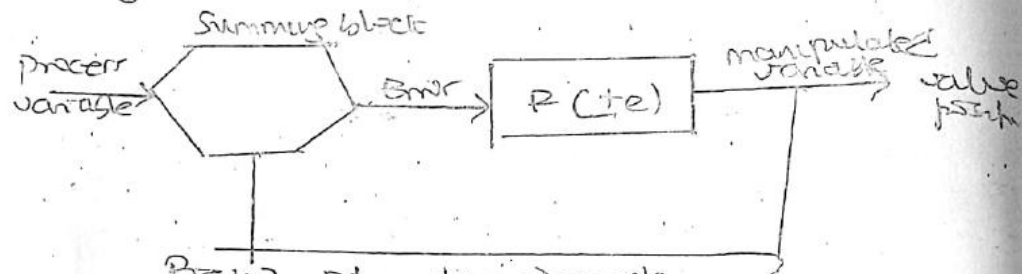


Fig 1.2 Algorithm example

The control algorithm can be used to calculate the requirements of much more complex loops.

ii) OFF SET

This is the sustained or consistent deviation of the process variable from the setpoint.

iii) LOAD DISTURBANCE

A load disturbance is an undesired change in one of the factors that can affect the process controller, such as the introduction of a new substance into a process which can alter the process variables.

c) THREE COMPONENTS OF ERROR

- 1) Error magnitude
- 2) Duration of error
- 3) Rate of change of error

i) ERROR MAGNITUDE

The magnitude of the error is simply the deviation between the values of the setpoint and the process variable. The magnitude of error at any point in time compared to the previous error provides the basis for determining the change in error. The change in error is also an important value.

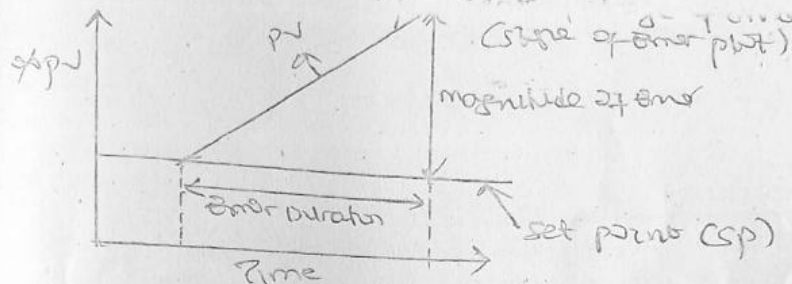
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DURATION OF ERROR?

Duration refers to the length of time that an error condition has existed.

RATE OF CHANGE OF ERROR

The rate of change of error is shown by the slope of the error plot shown below.



(4)

a) MANUAL SYSTEMS

In manual control systems, the control operations that involve human actions to effect necessary adjustments in a process.

AUTOMATIC SYSTEMS

Automatic control systems do not require human action; they require sensors and controllers using appropriate control algorithms to effect necessary adjustment in a process.

PRIMARY ELEMENT IN CONTROL SYSTEMS

Sensing devices are the first elements or primary elements in a control loop because they measure the process variables.

Examples of primary elements

Thermocouple

Displacement

Resistance temperature detector (RTD)

Ultrasonic emitter and receiver

Vortex shedder.

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- G) DP - Differential pressure
- RTD - Resistance Temperature Detector
- HVAC - Heating, ventilation and Air conditioning systems
- LPU - Lower Range Value

(i) OPERATION OF RTD

This is also known as a Resistance Temperature Detector, it is an electronic device used to determine the temperature by measuring the resistance of an electrical wire. This wire is referred to as the 'temperature sensor'. If the temperature is to be measured with high accuracy, an RTD is used. As the temperature of a process fluid surrounding the RTD rises or falls, the electrical resistance of RTD increases or decreases in a proportional amount. Hence, the resistance is measured, from which the temperature is determined.

⑤

(a) i) TRANSDUCERS

A transducer is a device that translates a mechanical signal into an electrical signal e.g. in a capacitance pressure device, transducer translate changes in pressure into a proportional change in capacitance.

ii) CONVERTER

A converter is a device that converts one type of signal into another type of signal, for example, in a process control loop, a converter was used to convert a 4-20 mA current signal into a 3-15 psig pressure signal for valve actuation - such a converter is called a current-to-pressure converter.

iii) TRANSMITTER

A transmitter is a device that converts a reading from a sensor or transducer into a standard signal and transmits it.

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signal to a controller or monitor.

Types of Transmitters

- 1) pressure transmitter
- 2) flow transmitter
- 3) temperature transmitter
- 4) Level Transmitter
- 5) Analytical Oxygen (O_2), carbon monoxide (CO), and pH Transmitter.

ii) SIGNAL

is a means of transmitting process variables measured from the instrument to a centralized control system.

Types of Signals in a Control System

- Pneumatic signal (pressure or fluid transmission)
- Analog signal (discrete, physical, tangible)
- Digital signal

Pneumatic Signals

are signals produced by changing the air pressure in a signal pipe in proportion to the measured change in a process variable.

The common industry standard pneumatic signal range is 3-15 psig. The 3 corresponds to the lower range value (LV) and the 15 corresponds to the upper range value (UV).

Electrical Signals

are signals produced by generating and transmitting electrical current or voltage signals. Standard electrical signals include the 4-20 mA, 1-5 V and the pulse output. The current signal is a kind of gauge in which the 4 mA and 20 mA is the LV and UV.

For example, if an RTD and transmitter are installed in a process vessel, the transmitter goes to produce a current signal as the sensor's resistance properly changes in response to changes in temperature. The current signal

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is then sent to a control device, such as a burner, to value.

DIGITAL SIGNALS

Digital signals are discrete values that are generated and combined in specific ways to represent process variables. The methodology used in combining the digital signals is referred to as protocol.

(A)

(A) PROTOCOL

This refers to the methodology used in combining the digital signals ^{and} is referred to in two categories

(i) CATEGORIES OF PROTOCOLS

- (1) Open digital protocols
- (2) Proprietary digital protocols

PROPRIETARY DIGITAL PROTOCOLS

These are digital protocols owned by specific companies and can only be used with their permission. Eg. the vendor's Transport protocol and Sledge protocol.

OPEN DIGITAL PROTOCOLS

As the name implies, are protocols open to all companies. Such as Modbus protocol, Device Net protocol, and High-speed addressable remote transducer protocol (HART).

(B) INDICATOR

An indicator is a human-readable device that displays information about a process. Indicators may be as simple as pressure or temperature gauge or more complex such as a digital read-out device.

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recorder

recorder is a device that records the output of a measurement device - e.g. a chart recorder

controller

controller is a device that receives data from a measuring instrument, compares that data to a pre-determined set point, and if necessary signals a control element to take corrective action. e.g. PLC (programmable logic controller).

vacuum

is a space in which there is no matter or in which the pressure is so low that any particles in the space do not affect any processes being carried on there, it is a condition well below normal atmospheric pressure and is measured in units of pressure (the Pascal).

final control element

final control element is a component of the control system that acts to physically change the manipulated variable. Examples are actuators (valves), pump motors, solenoid.

$$p = \rho gh \quad \text{--- (I)}$$

$$V = \rho g \quad \text{--- (II)}$$

$$\therefore p = \rho gh \quad \text{--- (III)}$$

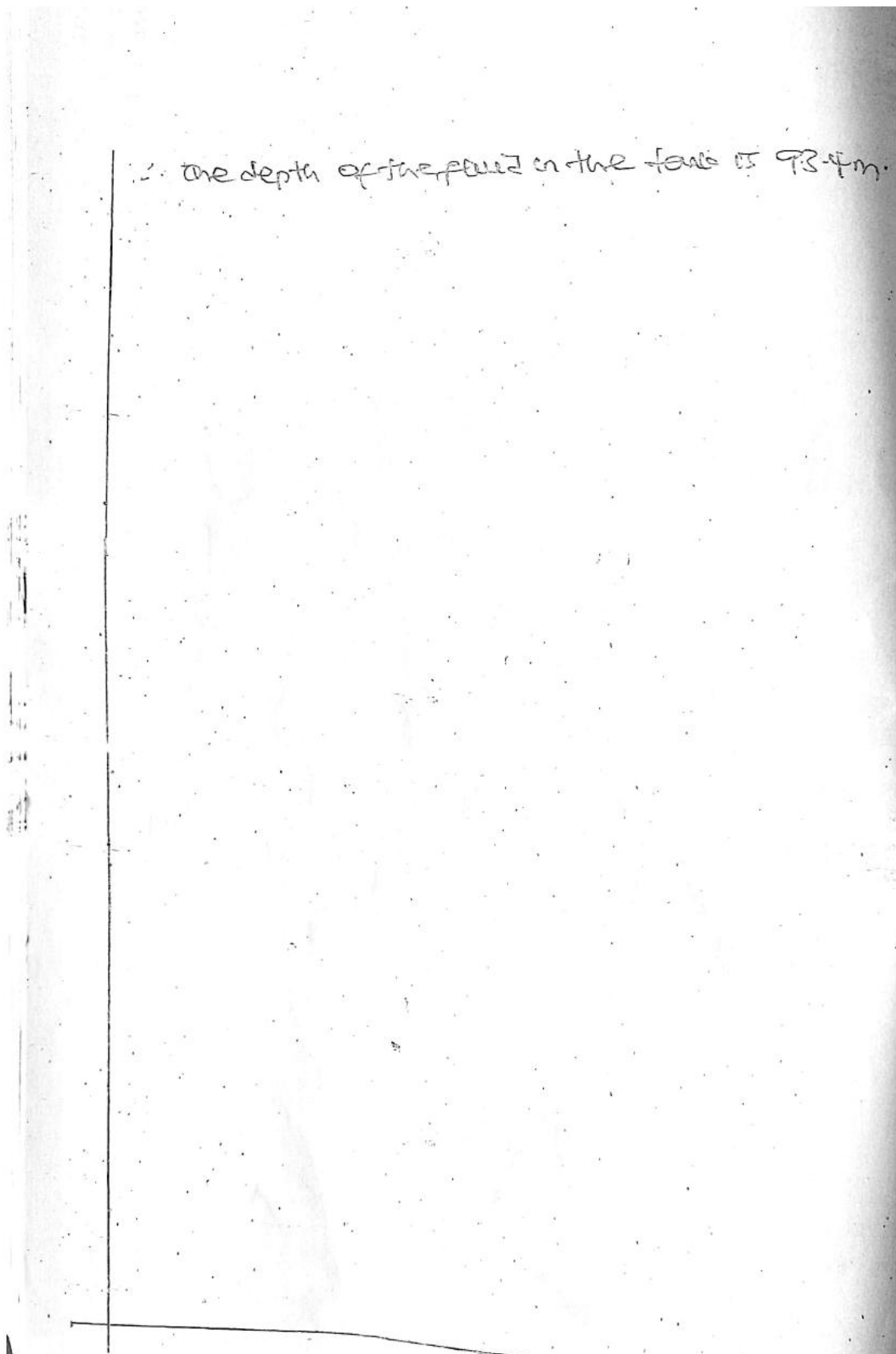
$$\therefore h = p / \rho g$$

$$h = \frac{1.27 \text{ MPa}}{13.6 \text{ kN/m}^3} = \frac{1.27 \times 10^6}{13.6 \times 10^3}$$

$$\text{NB } \text{N/m}^2 = \text{Pa}$$

$$\therefore h = 93.4 \text{ m}$$

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COLLEGE OF ENGINEERING TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING
FIRST SEMESTER EXAMINATIONS, 2020/2021 ACADEMIC SESSION

COURSE: (B. Eng. CHEMICAL ENGINEERING)
TITLE: CHEMICAL PROCESS DESIGN I
November 26, 2021 TIME: 2.30 – 5.00 pm

LEVEL: 400
COURSE CODE: CHE 413
DURATION: 2½ Hours

INSTRUCTIONS: Attempt any Five (5) questions. Brief and concise answers will be appreciated.

Q1

In addition to fixed capital investment costs associated with plant design, why should working capital be estimated and provided for, during the design stage?

Scaling is an important aspect of project evaluation, how is this accomplished?

The capital cost of a plant can be related to capacity by the equation $C_2 = C_1 \left(\frac{S_2}{S_1} \right)^n$ where C_1 and C_2

are the capital costs for plants of capacities S_1 and S_2 respectively. If the cost of the design and construction of a 200,000 litres/day sachet water plant was NGN10.56 million. What will be the capacity of a similar plant with a capital investment of NGN34.89 million. Take $n = 0.82$.

Q2

When designing a new chemical plant, one of the main decisions is the plant location. What should you bear in mind in choosing a suitable site? List six (6) factors to consider when choosing a site for a chemical plant.

What should be sited first when roughing out a plant layout? List six (6) principal factors to consider when roughing out a plant layout.

What techniques are usually used in site and plant layout by designers?

Q3

Why should process plant designers give serious attention to noise pollution and how can this be done?

How a diagrammatic representation of layers of plant safety, how does it work?

What is an intrinsically safe process, and what can lead to inherently unsafe conditions?

Q4

What will be the first consideration when designing a distillation column? What are other things to consider?

What separation correlation method is usually used in the design of columns for binary separation? List the steps involved in the design of distillation columns.

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Question 5

- (a) What is a process flowsheet?
- (b) Define or explain the following:
 - (i) Block flow diagram (BFD)
 - (ii) Fixed capital investment
 - (iii) Environmental audit
 - (iv) Design constraints
- (c) Why is the design of heat exchanger done on a trial and error basis?
- (d) What is the role of chemical engineers in the design of process vessels?

Question 6

- (a) What is a P & I diagram?
- (b) Describe what is usually included in a P & ID
- (c) Differentiate between a process flowsheet and a P & ID
- (d) Give the symbols of the following equipment:
 - (i) globe valve (ii) 3-way valve (iii) rotary motor (iv) fails open valve (v) restriction orifice

Question 7

- (a) What is basic office cost in process design and what does it include?
- (b) What are contingency charges and what does it cover?
- (c) What are offsite cost investments? List five (5) of its constituents

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CHE 413 (2020/2021)

1. (a) Working capital should be estimated and provided for during the design stage because the working capital is needed in maintaining plant operation. It is tied up in maintaining inventories of feeds, products and spare parts. It is also the difference between money owed by customers (accounts receivable) and money owed to suppliers (amount payable).

b) Scaling is accomplished by estimating the known cost of an earlier plant that used the same technology as from published data.

c) Data given:

$$S_1 = 200,000 \text{ litres/day} \quad C_1 = \text{₦ } 10.56 \text{ million } (10.56 \times 10^6)$$

$$S_2 = ? \quad C_2 = \text{₦ } 34.89 \text{ million } (34.89 \times 10^6)$$

$$n = 0.82$$

$$\text{Using the formula: } C_2 = C_1 \left[\frac{S_2}{S_1} \right]^n$$

making S_2 the subject of formula from the equation above

$$S_2 = S_1 \left[\frac{C_2}{C_1} \right]^{1/n}$$

$$S_2 = 200,000 \left[\frac{34.89 \times 10^6}{10.56 \times 10^6} \right]^{1/0.82}$$

$$S_2 = 858,116.8224 \text{ litres/day}$$

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29 In choosing a location for a suitable site one should consider the marketing area

- Factors to consider when choosing a site for a chemical plant

a) Raw material supply (b) Transport facilities

c) Location (d) Availability of labour

e) Availability of utilities (water, fuel, power)

f) Availability of suitable land (g) Climate

h) Environmental impact including effluent disposal

i) Local community considerations

j) Political, economic and strategic considerations

b) The economic construction and efficient operation of a process unit should be studied first when roughing a plant layout

Principal factors to consider when roughing a layout

i) Economic considerations (constructing and operating costs)

ii) The process requirements (iii) Convenience of operation

iv) Convenience of maintenance (v) Safety

vi) Future expansion (vii) Modular construction

c) Techniques used in site and plant layout by designers are

i) Cardboard cutouts of the equipment outlines

ii) Simple models made up from rectangular and cylindrical blocks

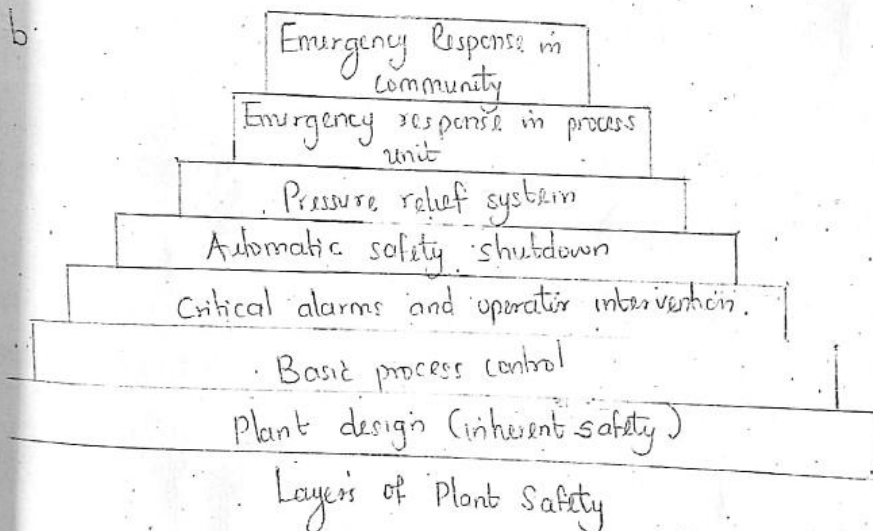
iii) Computer aided design (CAD) tools and computer models

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3a. Process plant designers give serious attention to noise pollution because noise causes a serious nuisance in the neighborhood of a process plant.

Attention is given to this problem of noise pollution by:

- Carefully selecting and specifying noisy equipments, ensuring that the vendor's specifications is checked to comply with statutory noise level.
- Noisy equipment should be fitted with silencers.
- They should as far as practicable be sited well away from the site boundary.



How it works: Each of the layers can be activated if the lower levels have all failed.

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C. An intrinsically safe process is one in which safe operation is inherent in the nature of the process i.e. a process that causes no danger or negligible danger under all foreseeable circumstances.

What can lead to inherently unsafe condition?

This occurs when the process conditions deviate from the design values.

4a. Determining the number of stages required for the desired degree of separation and the location of the feed tray is the first consideration when designing a distillation column.

Other things to consider are:

Tray spacing, column diameter, internal configurations, heating and cooling duties.

b. The McCabe-Thiele Separation Correlation method is usually used in the design of columns for binary separation.

Steps involved in the design of distillation columns:

- i) Specify the degree of separation required: set product specifications
- ii) Select the operating conditions: batch or continuous, operating pressure
- iii) Select the type of contacting device: plates or packing
- iv) Determine the stages and reflux requirements: The number of equilibrium stages
- v) Size the column: diameter, number of real stages
- vi) Design the column internals: plates, distributors, packing support

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vii) Mechanical designs: vessel and internal fittings

5a) A process flowsheet shows the arrangement of the major pieces of equipment and their interconnection. It is a description of the nature of the process.

b(i) Block Flow diagram (BFD): These are very simple diagrams composed of rectangles and lines indicating major materials or energy flow.

ii) Fixed Capital Investment is the total cost of designing, constructing and installing a plant and the associated modifications needed to prepare the plant site.

iii) Environmental audit: This is a systematic examination of how a business operation affects the environment. It includes all emissions to air, land and water, and covers the legal constraints, the effect on community, the landscape and the ecology.

iv) Design constraints: These are the control and limitations to the possibility of a design. It may either be external or internal. The internal constraints are those over which the designer exercises a measure of control while the external constraints are those outside the designer's control.

c) The design of a heat exchanger is done on a trial and error basis because the physical layout of the exchanger cannot be determined until the area is known.

d) The chemical engineer will be responsible for developing and specifying the basic design information for the vessel and also dictates the limits on the process conditions.

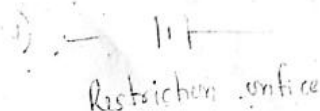
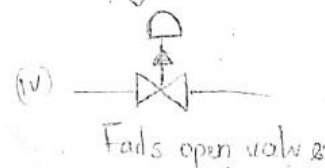
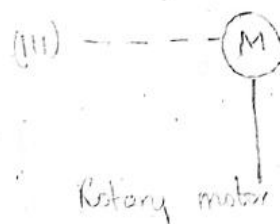
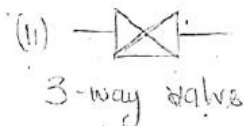
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Piping and Instrumentation diagram (P&ID) is a diagram that shows the engineering details of the equipment, instrument, piping, valves and fittings and their arrangement. It is often called the engineering flowchart or engineering line diagram.

b. The P&ID includes:

- i. All process equipment identified by an equipment number
- ii. All pipes identified by a line number (The pipe size and material of construction should be shown)
- iii. All valves control and block valves, with an identification number
- iv. Ancillary fittings that are part of the piping system
- v. Pumps, identified by a suitable code number
- vi. All control loops and instruments with an identification number

c. The process flowsheet shows the arrangement of the major pieces of equipment and their interconnection while the P&ID shows the engineering details of the equipment.



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(a) Home office costs also known as engineering costs or contractor charges, includes the costs of detailed design and other engineering services required to carry out the project.

It includes the cost of:

- Detailed design engineering of process equipment, piping systems, control systems and offshore, plant layout, drafting, cost engineering, scale models and civil engineering
- Procurement of main plant, tools and materials
- Construction supervision and services
- Administrative charges, including engineering supervision, project management, expediting, inspection, travel and living expenses and home office overheads
- Bonding — Contractor's profit

b) Contingency charges are extra costs added into the project budget to allow for variation from the cost estimate. It can be thought of as an additional fee charged by the engineering, procurement and construction (EPC) company to address the risk that the project will run over budget.

Contingency cost covers:

- Errors in the cost estimate
- Minor changes in project scope
- Labour disputes
- Other unexpected problems
- Changes in prices
- Currency fluctuations
- Subcontractor problems

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF CHEMICAL ENGINEERING
First Semester Examination; 2020/2021 Session

COURSE TITLE: Particulate System Analysis

COURSE CODE: CHE 415

INSTRUCTION: Answer Any Five Questions

Credit Units: 3

Time Allowed: 3 Hours

1. (i) State two (2) ways of generating centrifugal field in centrifugal separation
- (ii) List two (2) advantages of centrifugal separation
- (iii) Enumerate the mechanisms of size reduction in Chemical and process industry.
- (iv) A centrifuge is 5 ft in diameter and rotates at 50 Hz. At what frequency should a laboratory centrifuge of 0.5 mm diameter run, if it is to duplicate the plant condition?
- (v) In a centrifugal separation using a centrifuge, particles of density $2.5 \times 10^3 \text{ lb/ft}^3$ and spherical diameter size 9 mm were separated from suspension in water with a capacitive factor of 1000 m^2 . Calculate the throughput rate. Estimate the size cut corresponding for a suspension of coal particles in oil fed at the $0.03 \text{ cm}^3/\text{min}$. The coal density $13,500 \text{ kg/m}^3$ and oil of density 900 lb/ft^3 and its viscosity is 1 mNS/m^2 (Assume Stoke's law is applicable).
2. (i) Enumerate three factors that affect rate of sedimentation
- (ii) Show that $p = \frac{1}{2} \rho \omega^2 (R^2 - r_o^2)$ in a centrifugal separation
- (iv) A centrifuge has a basket which is 600 mm long and an inside diameter 100mm. is used to separate crystals from a dilute aqueous mother liquor. The optimum speed of rotation for the centrifuge is 50,000 rpm and the discharge weir is adjusted so that the liquid surface is 40mm from the basket wall. Assuming the crystals to be spherical and NONE smaller than $4\mu\text{m}$. determine
 - (i) The separating factor or effect (ii) Show that Stoke's law is valid (iii) The maximum flow rate of the mother liquor that can be processed by this centrifuge if all the crystals have to be removed.
 Data: Density of mother liquor = 990 kg/m^3 ; viscosity of mother liquor = $1.0 \times 10^{-3} \text{ kg/ms}$; density of crystal = 1150 kg/m^3 .

3a(i) Define the term porosity of a bed

(ii) Find the falling velocity of a spherical particle, 5 ft diameter, settling in a liquid having a viscosity and relative density of 5×10^{-3} Ns/m² and 0.90 respectively. The relative density of the solid is 3.6. Assume that the drag force on the particle is given by the Stoke's equation, $F = 3\pi\mu u$.

b(i) State two (2) factors that affect minimum fluidization velocity

(ii) Crude oil of density 950 kg/m³ and viscosity 6 mNs/m², is passed vertically upwards through a bed of catalyst consisting of approximately spherical particles of diameter 1×10^{-4} ft and density 2560 kg/m³. The particle voidage is assumed to be 0.5. At what approximate mass flow rate of flow per unit area of bed will (1) fluidization, and (2) transport of particles will occur.

4 a(i) List three (3) ways of characterizing particle in fluid particulate analysis.

(ii) A sample of dust from the air in a factory is collected on a glass slide. If dust on the slide was deposited from one cubic centimeter of air, estimate the mass of dust in g/m³ of air in the factory, given the number of particles in the various size ranges to be as follows:

Size range (μm)	0 - 1	1 - 2	2 - 4	4 - 6	6 - 10	10 - 14
Number of particles (-)	1500	1000	750	550	100	40

It is assumed that the particle is spherical and the density of the dust is 2550 kg/m³, and an appropriate allowance should be made for particle shape.

b(i) A mixture of limestone and galena of a size range from 0.020 ft to 0.070 ft is to be separated into two pure fractions using a hindered settling process. What will be the apparent minimum density of the fluid that will give this separation? The density of galena is 7400 kg/ft³ and the density of limestone is 2600 kg/ft³.

5a(i) In a water treatment plant the velocity gradient required for coagulation is 30 sec⁻¹, the volume of water in the mixing tank is 1000 dm³ with water viscosity of 40 Ns/m². Estimate the power input required to achieve coagulation.

(ii) Define hydrophilic and hydrophobic colloids

(iii) State two examples for each of the colloids type in (ii) above

(iv) Differentiate between flocculation and coagulation

b(i) Write briefly on the mechanism of operation of an electrostatic precipitator

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- (ii) Enumerate four (4) applications of an electrostatic precipitator in fluid particulates separation
 (iii) An electrostatic precipitator is to be constructed to remove fly ash particles from stack gases. the drift velocity of the fly ash is 30.000 m/s with 150 cm² as an area of the plate. Estimate each stack gas flow rates if the efficiency of the electrostatic precipitator is assumed to be 90% and 99%.

- 6 (a) A slurry containing 10 kg of water per kg of solid is to be thickening into a sludge containing 2.5 kg of water per kg of solid in a continuous operation. Laboratory test using five different concentration of the solid yielded the following results

Concentration (kg H ₂ O/kg solid)	7.0	6.2	5.7	5.1	4.5
Rate of sedimentation (m/s)	0.0002	0.00012	0.000094	0.00007	0.00005

Calculate the minimum area of the thickener to effect the separation of 15.5 kg of solid/seconds.

- (b) The size distribution of a particle is as measured by a microscope is as follows. Convert these data to obtain the distribution on a mass basis, and calculate the specific surface, assuming spherical particles with density 2600 kg/m³.

Size range (μm)	0-4	4-6	6-8	8-10	10-12	12-14	14-16
Number of particles in range (-)	1000	600	120	35	15	5	2

- 7a(i) What do you understand by the term filtration as a method of separation?

- (ii) Enumerate four factors to be considered in selecting equipment and operating conditions of a filter.

- ✓ (iii) Give two (2) advantages of filter press and one disadvantage of the filter press

- b(i) State three factors to be considered in filtration as a unit operation

- (ii) Filtration of an aqueous slurry is carried out initially at a constant rate of 200 cm³/s. After 140 Secs of filtration, the pressure drop developed across the filter is 5.50 kPa, and after 10mins, the pressure drop is 9.47 kPa. Calculate:

- (1) How long it would take to reach a pressure drop of 18 kPa
 (2) The volume of the filtrate which will have been collected at this stage

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CHE 415 (2020/21)

1. a (i) Two ways of generating centrifugal field in centrifugal separation

- Introduction of fluid with a high tangential velocity into a cylindrical or conical vessel
- Introduction of fluid into a form of rotating bowl and then accelerating rapidly at a constant angular velocity, ω and a forced vortex flow is established

(ii) Advantages of centrifugal separation

- High rate of separation can be easily achieved
- Separation of suspension and emulsion that are stable in gravitational field

(iii) Mechanism of size reduction

- Size reduction by impact
- Size reduction by attrition
- Size reduction by compression
- Size reduction by shear force produced by the fluid or particle.

b (i) Plant

$$d = 5\text{ft} \rightarrow 1.524\text{m}$$

$$f = 50\text{Hz}$$

$$r = 0.762\text{m}$$

$$\omega = 2\pi f = 2\pi(50) = 100\pi\text{rad/s}$$

For duplication to occur between laboratory and plant; the separating power (S.P) of both must be equal

Laboratory

$$d = 0.5\text{mm} \rightarrow 5 \times 10^{-4}\text{m}$$

$$r = 2.5 \times 10^{-4}\text{m}$$

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$$S.F = \frac{r\omega^2}{g}$$

$$\left(\frac{r\omega^2}{g} \right)_{\text{plant}} = \left(\frac{r\omega^2}{g} \right)_{\text{lab}}$$

$$= \frac{0.762 \times (100\pi)^2}{g} = \frac{2.5 \times 10^{-4} \omega^2}{g}$$

$$\omega_{\text{lab}}^2 = \frac{0.762 \times (100\pi)^2}{2.5 \times 10^{-4}} = 300825542.1$$

$$\omega_{\text{lab}} = \sqrt{300825542.1} = 17344.32305$$

$$\omega_{\text{lab}} = 2\pi f = 17344.32305$$

$$f_{\text{lab}} = \frac{17344.32305}{2\pi} = 2760.43 \text{ Hz}$$

(ii) Data given

$$\rho_s = 2.5 \times 10^3 \text{ lb/ft}^3 = 4 \times 10^4 \text{ kg/m}^3, \quad \mu = 0.001 \text{ Ns/m}^2$$

$$d = 9 \text{ mm} = 9 \times 10^{-3} \text{ m}$$

$$\rho = 1000 \text{ kg/m}^3$$

$$\Sigma = 1000 \text{ m}^2$$

Assuming Stoke's law: $\frac{d^2(\rho_s - \rho)g}{18\mu} = u_0$

$$u_0 = \frac{(9 \times 10^{-3})^2 (40000 - 1000) \times 9.81}{18 \times 0.001} = 1721.655 \text{ m/s}$$

$$\text{Throughput rate, } Q = u_0 \Sigma = 1721.655 \times 1000 = 1721655 \text{ m}^3/\text{sec}$$

$$\text{With oil feed, } Q = 0.08 \text{ cm}^3/\text{min} = 1.33 \times 10^{-9} \text{ m}^3/\text{sec}$$

$$\text{oil density, } \rho_o = 18,500 \text{ kg/m}^3$$

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$$\rho_{oil} = 900 \text{ lb/ft}^3 = 1.44 \times 10^4 \text{ kg/m}^3$$

$$U_0 = \frac{d^2 (\rho_c - \rho_{oil}) g}{18 \mu}$$

$$\text{To get } U_0 = \frac{Q}{\varepsilon} = \frac{1.33 \times 10^{-9}}{1000} = 1.33 \times 10^{-12} \text{ m/s}$$

$$\therefore 1.33 \times 10^{-12} = \frac{d^2 (18,500 - 14400) \times 9.81}{18 \times 0.001}$$

$$d^2 = \frac{1.33 \times 10^{-12} \times 18 \times 0.001}{(18,500 - 14400) \times 9.81} = 5.952 \times 10^{-19}$$

$$d = 7.715 \times 10^{-10} \text{ m}$$

20 (i) Factors that affect the rate of sedimentation

- Temperature - Stirring -

$$(ii) F = \text{mass} \times \text{acceleration} = m \times \frac{v^2}{r}$$

$$v = \omega r$$

$$F = m \frac{(\omega r)^2}{r} = m \omega^2 r$$

In terms of pressure: $p = \rho g h$

acceleration due to gravity, $g = \omega^2 r$

$$p = \rho \omega^2 r h$$

$$h = r$$

$$p = \rho \omega^2 r^2$$

$$dp = \rho \omega^2 r^2 dr$$

$$\text{Pressure gradient: } \frac{dp}{dr} = \rho \omega^2 r^2$$

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Integrating both parts

$$\int dp = \int_{r_0}^R \int \omega^2 r^2 dr$$

$$P = \frac{1}{2} \int \omega^2 (R^2 - r_0^2)$$

b) $h = 600 \text{ m} = 0.6 \text{ m}$

Inside diameter = $100 \text{ m} = 0.1 \text{ m}$ (basket diameter) = $2R$
 optimum speed of rotation = $50,000 \text{ rpm}$.

$$1 \text{ rpm} = 0.016667 \text{ Hz} \text{ also } 1 \text{ rpm} = 0.1047198 \text{ rad/s}$$

Discharge wheel diameter = $40 \text{ mm} = 0.04 \text{ m} = r$

Density of mother liquor (ρ_L) = 990 kg/m^3

viscosity of mother liquor (μ) = $1.0 \times 10^{-3} \text{ kg/ms}$

density of crystals (ρ_C) = 1450 kg/m^3

Assumptions

All crystals are spherical and none smaller than $4 \mu\text{m}$

1) THE SEPARATING FACTOR OR EFFECT

$K_{av} = \frac{\omega^2 (R+r)}{2g}$ is given as the average separation factor of a centrifuge

where ω = angular velocity of a centrifuge rotor
 g = gravitational acceleration

R = radius of basket = $\frac{0.1}{2} = 0.05 \text{ m}$

r = discharge wheel radius = $\frac{0.04}{2} = 0.02 \text{ m}$

THE ANGULAR VELOCITY (ω) IN rad/s CAN BE CONVERTED IN

WAYS
 If $1 \text{ rpm} = 0.016667 \text{ Hz}$
 $\therefore 50,000 \text{ rpm} = 833.35 \text{ Hz}$

And $\omega = 2\pi f$

$$\omega = 2 \times 3.142 \times 833.35$$

$$|\omega = 5236 \text{ rad/s}|$$

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So as stated above, rpm ω also = 0.1047198 rad/s

$$\therefore \text{So, } \omega \text{ rpm} = 50, \omega \times 0.1047198$$

$$\boxed{\omega = 5236 \text{ rad/s}}$$

$$\therefore K_{AV} = \frac{\omega^2 (R+r)}{2g} = \frac{5236^2 (0.05+0.02)}{2 \times 9.81}$$

$$K_{AV} = \frac{27415696 (0.07)}{19.62}$$

$$\boxed{K_{AV} = 97813.4}$$

to show that Stokes Law is valid

in a barrel radius of R m, and a radius of inner surface liquid r m and h m (the distance radially from the surface the liquid), the equation of motion in the radial direction a spherical particle of diameter (d m) under streamline motion is:

$$\frac{3}{6} (\rho_c - \rho_L) (r+h) \omega^2 - 3\pi \eta d u - \pi d^3 / 6 \rho_c \frac{du}{dt} = 0$$

where $3\pi \eta d u$ is the Stokes law (assuming the crystal to be small and none smaller than $4Hm$).

replacing u by dh/dt and neglecting the acceleration term as:

$$\frac{3}{6} (\rho_c - \rho_L) (r+h) \omega^2 - 3\pi \eta \frac{dh}{dt} - \pi d^3 / 6 \rho_c \frac{dh}{dt} = 0$$

$$\boxed{\frac{dh}{dt} = \frac{d^2 (\rho_c - \rho_L) \omega^2 (r+h)}{18\eta}} \rightarrow \text{shown}$$

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(ii) The maximum plate size of the mother liquor that can be processed by the centrifuge if all the crystals have to be removed.

The time of any element to remain in the basket is

$$V/\phi \quad \text{if } \phi = \phi_c \text{ (flow rate)}$$

$$\therefore V = \phi t$$

$$\text{and } t = V/\phi$$

where ϕ is the volumetric flow rate of feed to the centrifuge, V is the volume of the liquid retained in the basket at any t if the flow rate is adjusted so that a particle of diameter d is just retained, when it has to travel through the maximum distance (circumference) $h = CR - r$ before reaching the wall then:

$$h = \frac{d^2 (\rho_c - \rho_l) R \omega^2 V}{18 \mu \phi}$$

$$\text{And therefore } \phi = \frac{d^2 (\rho_c - \rho_l) R \omega^2 V}{18 \mu h}$$

$$\text{In this case, } V = \pi r^2 h \text{ or } \pi d^2 h$$

$$\therefore V = \frac{\pi}{4} (D^2 - d^2) h \text{ or } \pi (R^2 - r^2) h$$

$$V = \frac{\pi}{4} (0.1^2 - 0.04^2) \times 0.6$$

$$V = \frac{\pi}{4} (0.0084) \times 0.6$$

$$V = \frac{3.142}{4} \times 0.0084 \times 0.6$$

$$V = 0.00396 \text{ m}^3$$

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OPTIONAL CENTRIFUGATION PLASMA

$$\Phi = \frac{d^2 (\rho_c - \rho_L) R \omega^2 V}{18 \eta h}$$

where $d = 4 \mu\text{m}$ crystal diameter aren't smaller than $4 \mu\text{m}$

assumed $\therefore d = 4 \times 10^{-6} \text{ m}$

$$\rho_c = 1450 \text{ kg/m}^3$$

$$\rho_L = 990 \text{ kg/m}^3$$

$$R = 0.1 \text{ m}$$

$$\omega = 5236 \text{ rad/s}$$

$$V = 0.00396 \text{ m}^3$$

$$\eta = 1.0 \times 10^{-3}$$

and $h = \text{thickness of the centrifuge (crystal height)}$

$$h = \frac{(D-d)}{2} \text{ or } (R-r)$$

$$h = \frac{(0.1 - 0.04)}{2} \text{ or } 0.03 \text{ m}$$

$$\Phi = \frac{(4 \times 10^{-6})^2 (1450 - 990) \times 0.05 \times 5236^2 \times 0.00396}{18 \times 1.0 \times 10^{-3} \times 0.03}$$

$$\Phi = \frac{1.6 \times 10^{-11} \times 460 \times 0.05 \times 27415596 \times 0.00396}{18 \times 1.0 \times 10^{-3} \times 0.03}$$

$$\Phi = \frac{3.99528 \times 10^{-5}}{5.4 \times 10^{-4}}$$

$$\boxed{\Phi = 7.399 \times 10^{-2} \text{ m}^3/\text{s}}$$

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Hence the maximum flow rate of the mother liquor that
be processed by this Centrifuge if all the agitated
be removed is $Q = 7.799 \times 10^{-2} \text{ m}^3/\text{s}$

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3a(i) Porosity of a bed is defined as the measure of amount of space occupied by solvent (fluid) or solid support in a packed bed. It is a measure of the void spaces in a material.

$$ii) d = 5ft = 1.524m$$

$$\mu = 5 \times 10^{-5} \text{ N s/m}^2$$

$$R.D.L = 0.90 ; \text{Density} = 0.90 \times 1000 \text{ kg/m}^3 = 900 \text{ kg/m}^3$$

$$R.D.S = 3.6 ; \text{Density} = 3.6 \times 1000 \text{ kg/m}^3 = 3600 \text{ kg/m}^3$$

$$\text{From Stoke's equation, } F = \frac{3\pi \mu d u}{6} = \frac{\rho \pi d^3 g}{6}$$

$$u_t = \frac{d^2 (\rho_s - \rho) g}{18 \mu} = \frac{(1.524)^2 (3600 - 900) (9.81)}{18 \times 5 \times 10^{-5}}$$

$$= 68.35 \times 10^6 \text{ m/s}$$

b(i)-Size distribution

- Bed pressure drop

$$ii) \rho_{c.o} = 990 \text{ kg/m}^3$$

$$\mu_{c.o} = 6 \times 10^{-3} \text{ N s/m}^2$$

$$d = 1 \times 10^{-4} \text{ ft} = 3.048 \times 10^{-4} \text{ m}$$

$$\rho_s = 2560 \text{ kg/m}^3$$

$$e = 0.5$$

$$1. \frac{0.0055 e^3 (\rho_s - \rho) (d^2) g}{(1 - e) \mu}$$

$$= \frac{0.0055 \times (0.5)^3 (2560 - 990) (3.048 \times 10^{-4})^2 (9.81)}{(1 - 0.5) (6 \times 10^{-3})}$$

$$= 3.279 \times 10^{-4} \text{ m/s}$$

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Transport of particles occur when liquid velocity is equal to Particle terminal settling velocity (U_t) from Stokes law

$$U_t = \frac{d^2 g (\rho_s - \rho)}{18 \mu} = \frac{(3.048 \times 10^{-4})^2 (9.81) (2560 - 990)}{18 \times 6 \times 10^{-3}}$$

$$= 0.0132 \text{ m/s}$$

4 a(i) Composition, Size and shape

(5i) range	n	d	V(m ³)	mass of 1 particle	mass in size
0-1	1500	0.5	6.5×10^{-20}	1.69×10^{-13}	2.535×10^{-1}
1-2	1000	1.5	1.767×10^{-18}	4.51×10^{-12}	4.51×10^{-9}
2-4	750	3	1.414×10^{-17}	3.6×10^{-14}	2.7×10^{-8}
4-6	550	5	6.5×10^{-17}	1.669×10^{-10}	9.179×10^{-9}
6-10	100	8	2.68×10^{-16}	6.836×10^{-10}	6.836×10^{-10}
10-14	40	12	9.05×10^{-16}	2.307×10^{-9}	9.228×10^{-10}
					2.015×10^{-10}

$$V = \frac{\pi d^3}{6}$$

mass = Density $\times$ volume

$$\text{Density given} = 2550 \text{ kg/m}^3 \rightarrow \text{g/m}^3$$

$$= 2550 \times 10^3 \text{ g/m}^3$$

b(i) Galena (A)

Size 0.02 ft

ρ 7400 kg/ft³

Limestone (B)

0.07 ft

2600 kg/ft³

$$\frac{dB}{dA} = \left(\frac{\rho_A - \rho}{\rho_B - \rho} \right)^j$$

$j = 0.5$ (Stokes)

$j = 1$ (Newton's)

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Stokes

$$\frac{0.07}{0.02} = \left(\frac{7400 - \rho}{2600 - \rho} \right)^{0.5}$$

$$3.5 = ()^{0.5} \quad [\text{multiply by } 1/0.5]$$

$$12.25 = \frac{7400 - \rho}{2600 - \rho}$$

$$12.25(2600 - \rho) = 7400 - \rho$$

$$31850 - 12.25\rho = 7400 - \rho$$

$$24450 = 11.25\rho$$

$$\rho = 2173.3 \text{ kg/ft}^3$$

Newton's

$$\frac{0.07}{0.02} = \left(\frac{7400 - \rho}{2600 - \rho} \right)^1$$

$$3.5 = \frac{7400 - \rho}{2600 - \rho}$$

$$3.5(2600 - \rho) = 7400 - \rho$$

$$9100 - 3.5\rho = 7400 - \rho$$

$$1700 = 2.5\rho$$

$$\rho = 680 \text{ kg/ft}^3$$

5 a) Velocity gradient, $G = 30 \text{ s}^{-1}$ Volume, $V = 1000 \text{ dm}^3 = 1 \text{ m}^3$ Viscosity, $\mu = 40 \text{ N s/m}^2$ $P = ?$

$$G = \sqrt{\frac{P}{\mu V}}$$

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$$P = G \times 10^{-4} - (50) (40/11) = 500000$$

(ii) Hydrophilic colloids have affinity for water due to existence of water soluble groups

Hydrophobic colloids have little affinity for water due to the absence of water soluble groups.

(ii) Hydrophilic: proteins, soap, synthetic detergents.

Hydrophobic: amino, carboxyl, sulphonic, hydroxyl.

iv) Flocculation is a water treatment process following coagulation. Flocculation uses gentle stirring to bring the larger particles together so that they will form larger more settleable clumps called FLOCS.

Coagulation is a water treatment process that causes very small suspended particles to attract to one another and form larger particles.

b(i) An electrostatic precipitator removes dust or other finely divided particles from gas by charging the particles inductively with an electric field, then attracting them to highly charged collector plates. The process depends on two stages:

The suspension passes through an electric charge area where ionization of the gas occurs, the ions produced collide with the suspended particles and confer on them electric charge. The charged particles drift towards an electrode of opposite sign and are deposited on the electrode where their electric charge is neutralized.

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Paper 2 of 2 — CHE 411 SEPARATION PROCESS II 2023I2024 2020I2021

Level 400 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

First Semester Examination, 2023/2024 Session

COURSE TITLE: SEPARATION PROCESSES II

COURSE CODE: CHE 411

COURSE CREDIT UNIT: 3

INSTRUCTION: Answer Five (5) Questions: Question 1 and any other four. Start each question on a fresh page

Time Allowed: 3 Hours

SECTION A: COMPULSORY QUESTION

1. (a) State the significance of reflux ratio in a distillation column.
(b) A saturated liquid feed having a flowrate of 2,500 kmol/h, with a composition of 0.056-mole fraction of propane, 0.321 *n*-butane, 0.482 *n*-pentane, and the remaining *n*-hexane, is to be distilled in an atmospheric distillation column. The column has a total condenser and a partial re-boiler, with a reflux ratio of 3.5, and the reflux is a saturated liquid. A fractional recovery of 99.4% of *n*-butane is desired in the distillate, and the required fractional recovery of *n*-pentane in the bottoms is 99.7%. If the optimal feed stage is used,
(i) In the given distillation scenario, how would you identify the key components?
(ii) Estimate the distillate and bottom compositions and flow rates.

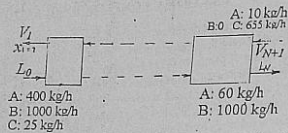
SECTION B: ANSWER ANY FOUR (4) QUESTIONS

2. (a) Draw and label a typical fixed-bed apparatus used in sugar beet leaching.
(b) In multicomponent gas absorption systems, designing solvent mixtures that leverage component interactions can significantly enhance mass transfer efficiency. Discuss in detail.

3. (a) List three factors you would weigh when deciding on an appropriate solvent for an absorption process.
 (b) What is the primary driving force behind reverse osmosis?
 (c) State how the membrane in reverse osmosis differs from a typical filtration membrane.
 (d) In what way does the design of the permeate collection system in a reverse osmosis setup affect the reliability of the water purification process?

4. (a) Evaluate, exhaustively, the importance of pilot plant testing in the validation of new separation processes before full-scale deployment.
 (b) Outline the role of academia and industrial Research & Development (R&D) centers in the emergence of novel separation processes. Give two points each.

5. A continuous countercurrent multistage system is to be used to leach oil from meal containing 400 kg oil (A) and also 25 kg benzene (C). The inlet flow per hour of the solvent mixture contains 655 kg benzene and 10 kg oil. The leached solids are to contain 60 kg oil. Settling experiments similar to those in the actual extractor shows that solution retained depends on the concentration of oil in the solution. The diagram is shown:



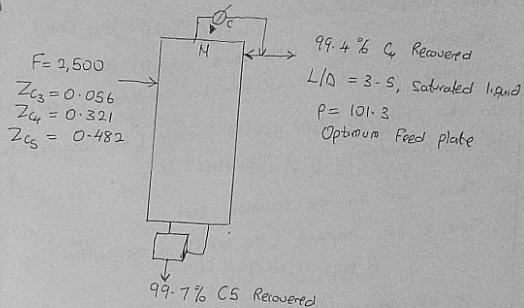
The data are tabulated below as N kg inert solid B /kg solution and y_A kg oil A /kg solution

N	2.00	1.98	1.94	1.89	1.82	1.75	1.68	1.61
y_A	0	0.10	0.20	0.30	0.40	0.50	0.60	0.70

- (i) Define the concept of 'countercurrent flow' in the context of this problem.
 (ii) Briefly explain the meaning of the word 'continuous' in the context of this problem.
 (iii) Determine the values of V_1 , x_1 , and L_N .

CHE 411: SEPARATION PROCESSES II 2023/2024

- 1
 a) Reflux ratio affects separation efficiency, column stability, energy consumption, and product quality, which influences the distillation column design, optimization and overall performance.



$$Z_{C_4} = 1 - (0.056 + 0.321 + 0.482)$$

$$= 1 - 0.859$$

$$= 0.141$$

- a) Of these 4 components, the order of volatility is as follows: propane > n-butane > n-pentane > n-hexane. Since the fractional recovery of n-butane (in the distillate) is specified, n-butane is the key component. Also, the fractional recovery of n-pentane (in the bottom) is specified, n-pentane is another key component.

Of these 2 key components (i.e. n-butane and n-pentane) n-butane is the most volatile species, while n-pentane is the least volatile one.

Thus,

- n-butane is the light key component (LK)
- n-pentane is the heavy key component (HK)

Hence, propane and n-hexane are the non-keys (NKS).

As propane is lighter (i.e. more volatile) than the LK (i.e. n-butane), it is the light non-key component (LNK).

On other hand, n-hexane is less volatile (i.e. heavier) than HK (i.e. n-pentane), it is the HNK.

b) Thus, we can be assumed that

- There is no propane (the LNK) in the bottom; i.e.

$$Bx_{C_3, bot} = 0$$
 or all propane (C_3) appears only in the distillate; i.e.

$$Dx_{C_3, dist} = Fz_{C_3}$$
- There is no n-hexane (the HNK) in the distillate; i.e.

$$Dx_{C_6, dist} = 0$$
 or all n-hexane (C_6) comes out only in the bottom; i.e.

$$Bx_{C_6, bot} = Fz_{C_6}$$

From the given information regarding fractional recovery, we write the following equations for n-butane C_4 and n-pentane C_5 .

$$Dx_{C_4, dist} = (0.994) Fz_{C_4}$$

$$Bx_{C_4, bot} = (1 - 0.994) Fz_{C_4}$$

$$Bx_{C_5, bot} = (0.997) Fz_{C_5}$$

$$Dx_{C_5, dist} = (1 - 0.997) Fz_{C_5}$$

We also have the following 2 equations:

$$\sum (Dx_{i, dist}) = D$$

$$\sum (Bx_{i, bot}) = B$$

Let's start the calculations with the distillate

$$Dx_{C_2, dist} = Fz_{C_2} = (2,000)(0.056) = 140$$

$$Dx_{C_6, dist} = 0$$

$$Dx_{C_4, dist} = (0.994) Fz_{C_4} = (0.994)(2,500)(0.321) = 797.685$$

$$Dx_{C_5, dist} = (1 - 0.997) Fz_{C_5} = (1 - 0.997)(2,500)(0.482) = 3.615$$

Hence

$$\sum (Dx_{i, dist}) = D = Dx_{C_2, dist} + Dx_{C_4, dist} + Dx_{C_5, dist} + Dx_{C_6, dist}$$

$$= D = 140 + 797.685 + 3.615 + 0 = 941.3$$

Doing the same for the bottom,

- $Bx_{C_3, bot} = 0$
- $R_{C_6, bot} = Fz_{C_6} - (2,500)(0.141) = 352.5$

$Bx_{4,bot} = (1 - 0.994) F Z_{C_4}$
 $= (1 - 0.994) (2,500) (0.321) = 4.815$
 $Bx_{C_5,bot} = (0.997) F Z_{C_5}$
 $= (0.997) (2,500) (0.482) = 1,201.385$
 $\therefore \sum (Bx_{i,bot}) = Bx_{C_5,bot} + Bx_{4,bot} + Bx_{3,bot} + Bx_{6,bot}$
 $= B = 0 + 4.815 + 1,201.385 + 352.5$
 $= 1,558.7$
 Check if $F = D + B$?
 $2,500 = 941.3 + 1,558.7 = 2,500$! ok!

This means that our first guess, with the assumptions that

- all LHK (i.e. propane in this example) goes out only in the distillate.
- all HHK (i.e. n-hexane in this example) appears only in the bottom

 seems to yield the satisfactory answers.

The compositions of each species in the distillate and the bottom can be computed as follows:

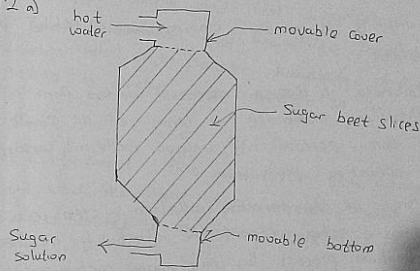
Distillate

$- C_3: x_{C_3,dist} = \frac{Dx_{C_3,dist}}{D} = \frac{140}{941.3} = 0.149$
 $- C_4: x_{C_4,dist} = \frac{Dx_{C_4,dist}}{D} = \frac{797.685}{941.3} = 0.847$
 $- C_5: x_{C_5,dist} = \frac{Dx_{C_5,dist}}{D} = \frac{3.615}{941.3} = 0.004$
 $- C_6: x_{C_6,dist} = \frac{Dx_{C_6,dist}}{D} = \frac{0}{941.3} = 0$

and
 $\sum x_{i,dist} = 0.149 + 0.847 + 0.004 + 0 = 1$

Bottom

$- C_3: x_{C_3,bot} = \frac{Bx_{C_3,bot}}{B} = \frac{0}{1,558.7} = 0$
 $- C_4: x_{C_4,bot} = \frac{Bx_{C_4,bot}}{B} = \frac{4.815}{1,558.7} = 0.003$
 $- C_5: x_{C_5,bot} = \frac{Bx_{C_5,bot}}{B} = \frac{1,201.385}{1,558.7} = 0.771$
 $- C_6: x_{C_6,bot} = \frac{Bx_{C_6,bot}}{B} = \frac{352.5}{1,558.7} = 0.226$
 and
 $\sum x_{i,bot} = 0 + 0.003 + 0.771 + 0.226 = 1$

2 a) 

A Typical fixed-bed apparatus for sugar beet leaching.

b) Optimized mixtures improve absorption capacity, selectivity and process economics. When designing effective solvent mixtures, solubility, diffusivity and reactivity are considered. This approach enables targeted absorption of specific gas components, reducing solvent usage and improving process performance. By optimizing solvent mixtures, industries can increase efficiency, reduce costs and improve environmental sustainability in gas absorption applications.

3

- a) i) Solubility
- ii) Volatility
- iii) Viscosity
- iv) Compressibility
- v) Cost.

b) Pressure is the primary driving force behind reverse osmosis.

c) The membrane in reverse osmosis differs from a typical filtration membrane in terms of its smaller pore size, semi-permeability, unique material composition and high operating pressure.

d) The design of the permeate collection system in a reverse osmosis setup affects the reliability of the water purification process by minimizing contamination risk, reducing pressure drops, preventing scaling and fouling, ensuring consistent flow rates, and facilitating maintenance and cleaning.

4

a) Process Validation and Optimization: Pilot plant testing allows for the validation of process design, simulation and modeling. It helps to identify potential bottlenecks, optimize process conditions, and refine process design.

ii) Scale-up and Commercialization: Pilot plant testing provides valuable data for scale-up and commercialization. It helps to determine the feasibility of the process, identify potential risks and develop strategies for mitigating them.

b) Fundamental Research and Development: Academia provides a platform for fundamental research and development of novel separation processes. Researchers can explore new ideas, develop theories, and test hypotheses.

ii) Talent Development and Knowledge Transfer: Academia develops talent and transfers knowledge to industry partners to develop and commercialize new technologies.

5

a) Counter-current flow refers to the flow arrangement where the solid feed moves in one direction and the liquid solvent moves in the opposite direction through multiple stages.

b) The word "continuous" refers to the uninterrupted flow of materials (solid and liquid) through the system, with no breaks or batches, allowing for a steady-state operation.

iii)

Informing given

Feed slurry (L_0)
 $A = 400 \text{ kg/h}$ $B = 1000 \text{ kg/h}$ $C = 25 \text{ kg/h}$

Entering solvent (V_{N+1})
 $A = 10 \text{ kg/h}$ $B = 0 \text{ kg/h}$ $C = 655 \text{ kg/h}$

Under flow solution
 $A = 60 \text{ kg/h}$ $B = 1000 \text{ kg/h}$ $C = ?? \text{ kg/h}$

Find coordinate at L_0
 mass of $A = 400 \text{ kg/h}$
 mass of $B = 1000 \text{ kg/h}$
 mass of $C = 25 \text{ kg/h}$

$$Y_{A0} = \frac{A}{L_0} = \frac{A}{A+B} = \frac{400}{400+25} = \frac{400}{425} = 0.94$$

$$N_0 = \frac{B}{L_0} = \frac{B}{A+B} = \frac{1000}{400+25} = \frac{1000}{425} = 2.35$$
 Coordinate for L_0 (Y_{A0}, N_0) = (0.94, 2.35)

Find coordinate at V_{N+1}
 mass of $A = 10 \text{ kg/h}$
 mass of $B = 0 \text{ kg/h}$
 mass of $C = 655 \text{ kg/h}$

from material balance calculations
 Total solution balance
 $L_0 + V_{N+1} = L_N + V_1 = m$
 $L_0 + V_{N+1} = m$
 $425 + 665 = m$ $m = 1090 \text{ kg}$

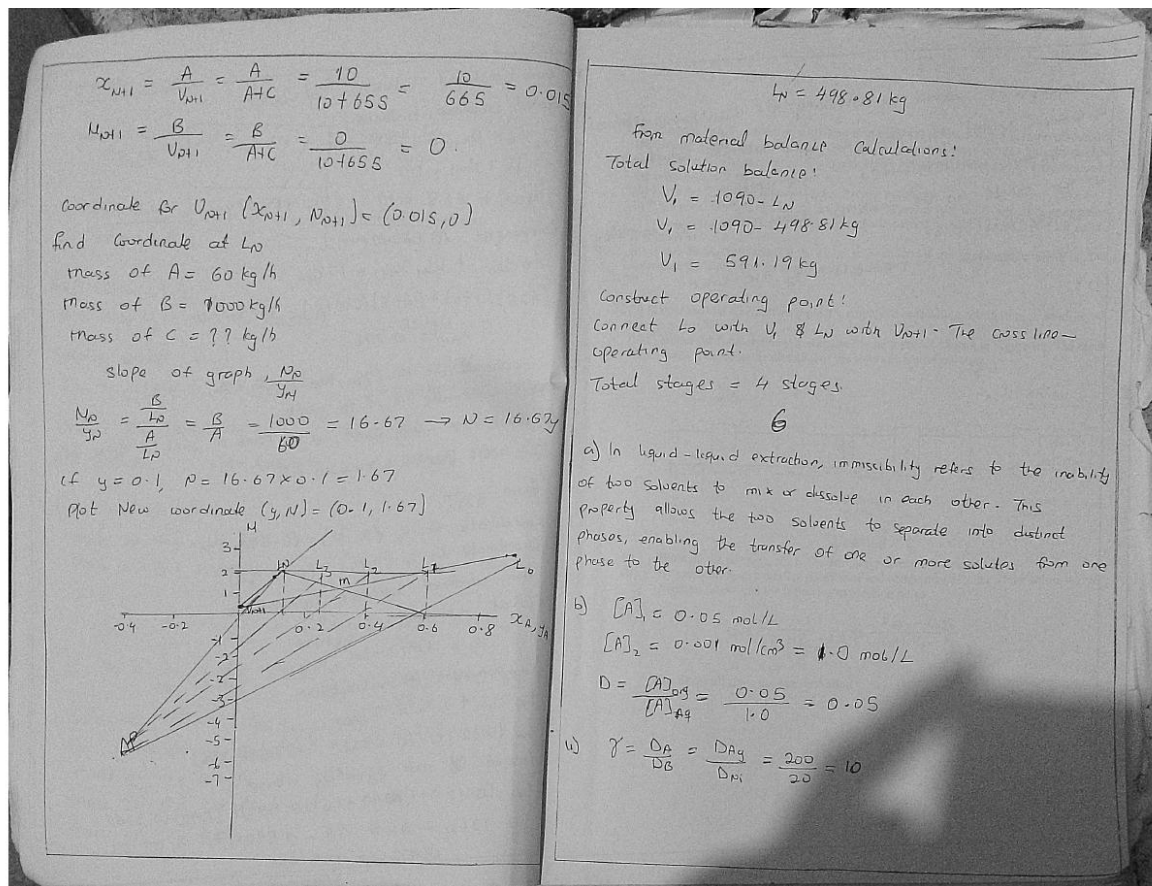
Component A balance
 $L_0 Y_{A0} + V_{N+1} X_{N+1} = m X_{Am}$
 $(425)(0.94) + (665)(0.015) = (1090) X_{Am}$
 $X_{Am} = 0.376$

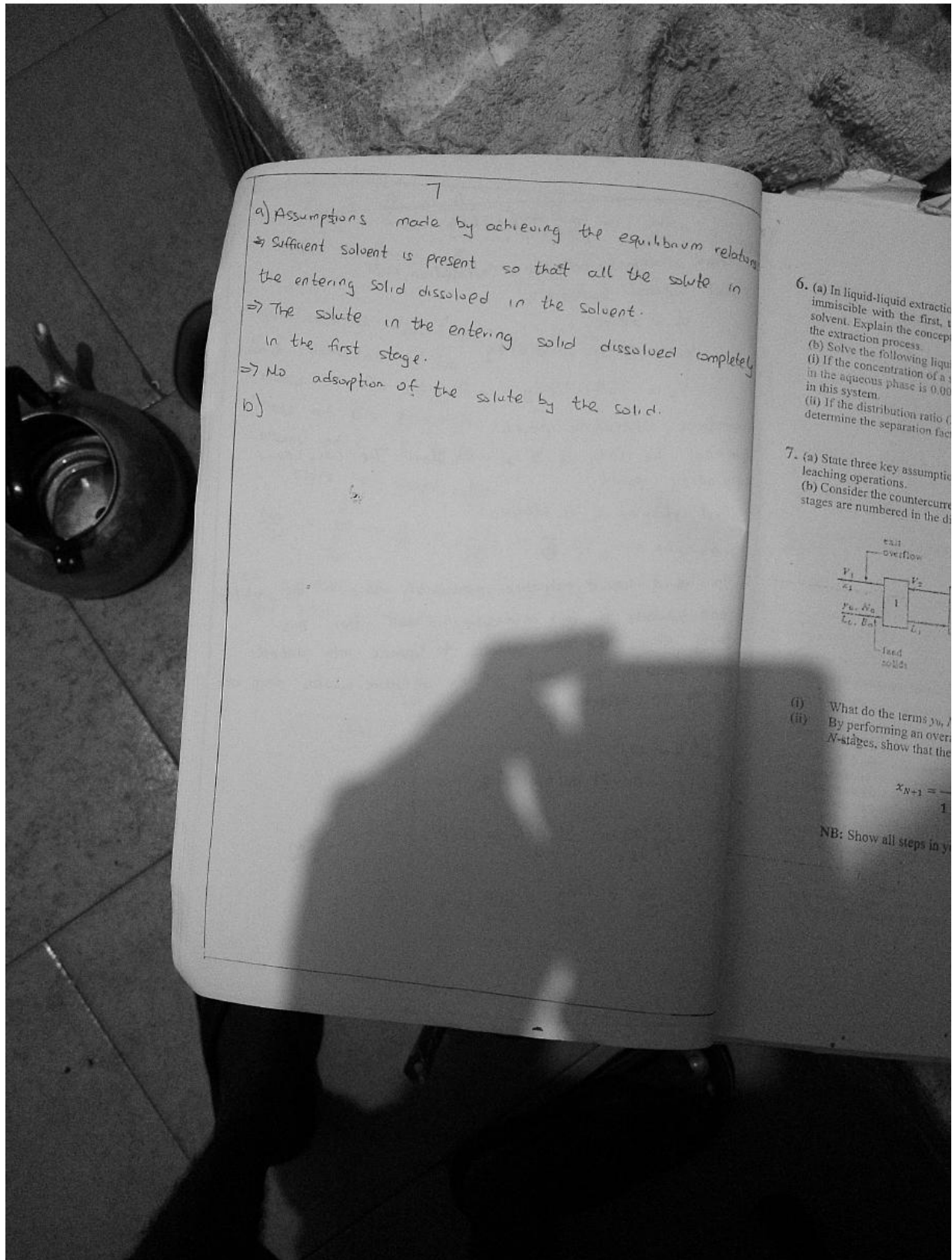
Coordinate for m (X_{Am}, N_{Am}) = (0.376, 0.916)
 Plot coordinate m in the graph
 Construct line from point L_0 to point m until it crosses at x -axis Point at x -axis = V_1

From graph,
 Coordinate for V_1 (X_1, N_1) = (0.592, 0)
 Coordinate for L_N (Y_N, N_N) = (0.12, 2.0)

from material balance calculations
 Total solution balance
 $V_1 = 1090 - L_N$ [1]

Component A balance
 $L_N Y_N + V_1 X_1 = m X_{Am}$
 $L_N (0.12) + V_1 (0.592) = (1090) (0.376)$
 Insert [1] into equation above
 $L_N (0.12) + (1090 - L_N) (0.592) = (1090) (0.376)$
 $0.12 L_N + 645.28 - 0.592 L_N = 410.96$
 $0.472 L_N = 235.44$





6. (a) In liquid-liquid extraction, a solution is brought into contact with a solvent, essentially immiscible with the first, to bring the transfer of one or more solutes into the second solvent. Explain the concept of this immiscibility between solvents and its significance in the extraction process.
- (b) Solve the following liquid-liquid extraction problems:
- (i) If the concentration of a solute in the organic phase is 0.05 mol/L and its concentration in the aqueous phase is 0.0001 mol/cm³, calculate the distribution ratio (D) for the solute in this system.
- (ii) If the distribution ratio (D) for nickel is 20 and the distribution ratio for silver is 200, determine the separation factor (γ) for the system.
7. (a) State three key assumptions typically employed in reaching equilibrium in solid-liquid leaching operations.
- (b) Consider the countercurrent multistage leaching process shown in Figure 1.0. The ideal stages are numbered in the direction of the solids or underflow stream.

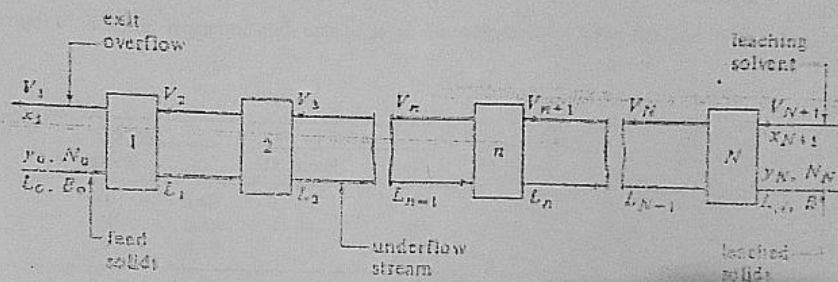


Figure 1.0

- (i) What do the terms y_0 , N_0 , L_0 , L_N and x_1 represent in the process?
- (ii) By performing an overall mass balance and component balance of the solute A over N -stages, show that the following expression holds:

$$x_{N+1} = \frac{1}{1 + \frac{(V_1 - L_0)}{L_N}} y_N + \frac{V_1 x_1 - L_0 y_0}{L_N + V_1 - L_0}$$

NB: Show all steps in your derivation.

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF CHEMICAL ENGINEERING

FIRST SEMESTER; 2020/2021 SESSION EXAMINATION

COURSE TITLE: SEPARATION PROCESSES II

Credit units: 3

COURSE CODE: CHE 411

Time Allowed: 3 Hours.

INSTRUCTIONS: Answer any five (5) Questions. All Questions carry equal marks

- 1) A mixture of hexane, heptane, and octane is to be separated to give the following products.

Component	Feed		Product		Bottoms		Relative volatility
	F (kmol)	x_F	D (kmol)	x_D	W (kmol)	x_W	
Hexane	40	0.40	40	0.534	0	0	2.70
Heptane	35	0.35	34	0.453	1	0.04	2.22
Octane	25	0.25	1	0.013	24	0.96	1.0

- a) What will be the value of the minimum reflux ratio, if the feed is liquid at its boiling point?
- b) Determine the number of theoretical plates required
- c) Using the attached chart (Figure 1), investigate the change in the number of plates with regards to the reflux ratio.
- 2) A mixture of ortho, meta, and para-methylnitrotoluenes containing 60, 4, and 36 mole per cent respectively of the three isomers is to be continuously distilled to give a top product of 98 mol per cent ortho, and the bottom is to contain 12.5 mole per cent ortho. The mixture is to be distilled at a temperature of 410 K requiring a pressure in the boiler of about 6.0 kN/m². If a reflux ratio of 5 is used, how many ideal plates will be required and what will be the approximate compositions of the product streams? The volatility of ortho relative to the para isomer may be taken as 1.70 and of the meta as 1.16 over the temperature range of 380–415K
- 3) 160 cm³/s of a solvent S is used to treat 400 cm³/s of a 10 per cent by mass solution of A in B, in a three-stage countercurrent multiple-contact liquid-liquid extraction plant.
- a) What is the composition of the final raffinate?

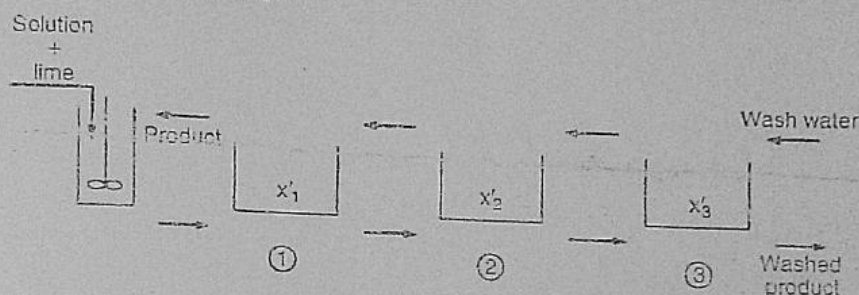
b) Using the same total amount of solvent, evenly distributed between the three stages, what would be the composition of the final raffinate if the equipment were used in a simple multiple-contact arrangement?

The Equilibrium data are as follows:

kg A/kg B:	0.05	0.10	0.15
kg A/kg S:	0.069	0.159	0.258
Densities (kg/m ³):	$\rho_A = 1200$,	$\rho_B = 1000$,	$\rho_S = 80$

4) a) In a pilot scale test using a vessel 1 m³ in volume, a solute was leached from an inert solid and the water was 75 per cent saturated in 100 s. If, in a full-scale unit, 500 kg of the inert solid containing, as before, 28 per cent by mass of the water-soluble component, is agitated with 100 m³ of water, how long will it take for all the solute to dissolve, assuming conditions are equivalent to those in the pilot scale vessel? Water is saturated with the solute at a concentration of 2.5 kg/m³.

b) Caustic soda is manufactured by the lime-soda process. A solution of sodium carbonate in water, containing 0.25 kg/s Na₂CO₃, is treated with the theoretical requirement of lime, and after the reaction is complete the CaCO₃ sludge, containing 1 part of CaCO₃ per 9 parts of water, by mass, is fed continuously to three thickeners in series and washed countercurrently, as shown in the following Figure.



Calculate the necessary rate of feed of neutral water to the thickeners so that the calcium carbonate, on drying, contains only 1 per cent of sodium hydroxide. The solid discharged from each thickener contains 1 part by mass of calcium carbonate to 3 of water. The concentrated wash liquid is mixed with the contents of the agitator before being fed to the first thickener.

5) An ultrafiltration plant is required to treat 50 m³/day of a protein-containing waste stream. The waste contains 0.5 kg/m³ of protein which has to be concentrated to 20 kg/m³ so as to allow recycling to the main process stream. The tubular membranes to be used are available as 30 m² modules. Pilot plant studies show that the flux J through these membranes is given by:

$$J = 0.02 \ln (30/C_f) \text{ m/h.}$$

where C_f is the concentration of protein in kg/m³. Due to fouling, the flux never exceeds 0.04 m/h. Estimate the minimum number of membrane modules required for the operation of this process as,

- a) a single feed and bleed stage,
 b) two feed and bleed stages in series.

Operation for 20 hr/day may be assumed.

6) You have been requested to redesign a water-cooling tower that has a blower with a capacity of 8.30×10^6 ft³/hr of moist air (at 80°F and a wet-bulb temperature of 65°F). The exit air leaves at 95°F and 90°F wet bulb. How much water can be cooled in pounds per hour if the water to be cooled is not recycled, enters the tower at 120°F, and leaves the tower at 90°F?

7) A single pellet of alumina is exposed to a flow of humid air at a constant temperature. The increase in mass of the pellet is followed automatically, yielding the following results:

$t(\text{min})$	2	4	10	20	40	60	120
$x_r (\text{kg/kg})$	0.091	0.097	0.105	0.113	0.125	0.128	0.132

Assuming the effect of the external film is negligible, predict time t against x_r values for a pellet of twice the radius, where x_r is the mass of adsorbed phase per unit mass of adsorbent.

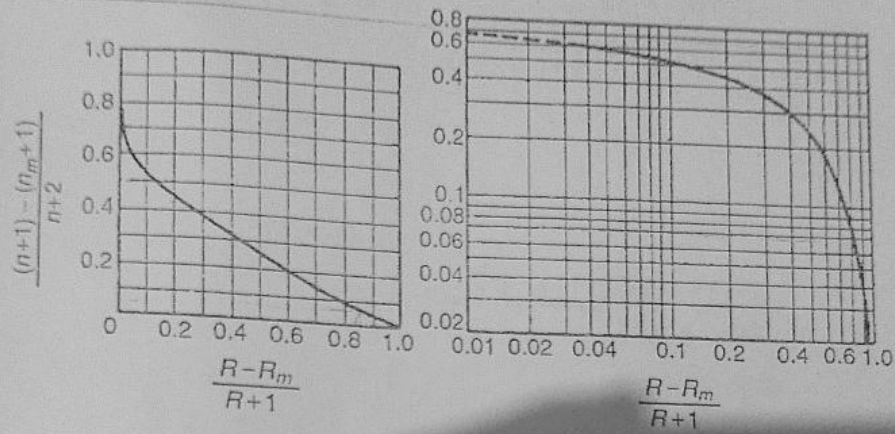
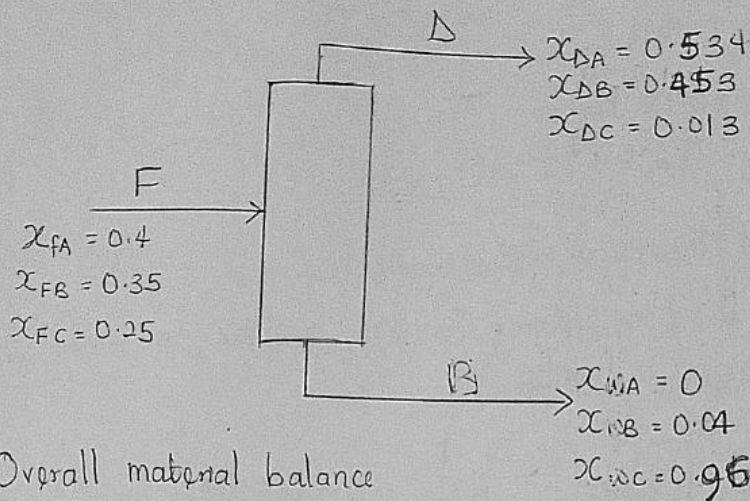


Figure 1: Relationship between Reflux ratio and Number of Plates

CHE 411 (2020/2021)

1



Overall material balance

$$F = D + B$$

a. Minimum reflux ratio when feed is liquid, R_m

Using the formula:

$$\frac{\alpha_A x_{FA}}{\alpha_A - \theta} + \frac{\alpha_B x_{FB}}{\alpha_B - \theta} + \frac{\alpha_C x_{FC}}{\alpha_C - \theta} + \dots = 1 - q$$

where $\alpha_A = 2.70$, $\alpha_B = 2.22$, $\alpha_C = 1.0$, $q = 1$ (feed is liquid)

$$\left(\frac{2.70 \times 0.4}{2.70 - \theta} \right) + \left(\frac{2.22 \times 0.35}{2.22 - \theta} \right) + \left(\frac{1 \times 0.25}{1 - \theta} \right) = 1 - 1 = 0$$

The required value of θ must satisfy the relation $\alpha_B < \theta < \alpha_A$
 that is $1.0 < \theta < 2.22$ where light key (A) is heptane and heavy
 key (B) is octane.

Assuming $\theta = 1.17$ (after trying values of θ between this range)

$$\text{then } \left(\frac{2.70 \times 0.4}{2.70 - 1.17} \right) + \left(\frac{2.22 \times 0.35}{2.22 - 1.17} \right) + \left(\frac{1 \times 0.25}{1 - 1.17} \right) = 0$$

To get the minimum ratio, R_m

$$\frac{\alpha_A x_{DA}}{\alpha_A - \theta} + \frac{\alpha_B x_{DB}}{\alpha_B - \theta} + \frac{\alpha_C x_{DC}}{\alpha_C - \theta} + \dots = R_m + 1$$

$$\left(\frac{2.70 \times 0.534}{2.70 - 1.17} \right) + \left(\frac{2.22 \times 0.453}{2.22 - 1.17} \right) + \left(\frac{1.00 \times 0.013}{1.00 - 1.17} \right)$$

$$= 1.827$$

$$1.827 = R_{n+1}$$

$$R_m = 1.827 - 1 = 0.827$$

b. Number of theoretical plates required.

This can be found by applying Fenske's equation to the key components (Heptane and Octane)

$$n+1 = \log \left[\left(\frac{x_A}{x_B} \right)_d \left(\frac{x_B}{x_A} \right)_w \right] \quad \alpha_{AB} = \frac{\alpha_A}{\alpha_B} = \frac{2.22}{1} = 2.22$$

n is the number of plates

$$n+1 = -\log \left[\frac{0.0453}{0.013} \times \frac{0.96}{0.04} \right] = \frac{\log 835}{\log 2.22} = 8.5$$

$$n+1 = 8.5 \quad ; \quad n = 8.5 - 1 = 7.5 \approx 8$$

Number of theoretical plates = 8

c. To investigate the change in number of plates, n with the reflux ratio, R

$$\text{Since } R_m = 0.8 \text{ and } n+1 = 8.5$$

$$\text{Using the chart in Fig 1 } \left(\frac{n+1 - (n_m+1)}{n+2} \right) \text{ vs } \frac{R - R_m}{R+1}$$

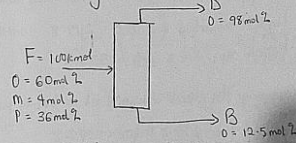
R	$\frac{R - R_m}{R + 1}$	$\frac{(n+1) - (n_m+1)}{n+2}$	n
1	$\frac{1 - 0.83}{1 + 1} = 0.085$	0.55	19
2	$\frac{2 - 0.83}{2 + 1} = 0.390$	0.30	11.8
5	$\frac{5 - 0.83}{5 + 1} = 0.695$	0.15	9.1
10	$\frac{10 - 0.83}{10 + 1} = 0.833$	0.08	8.2

NB: The values from $\frac{(n+1) - (n_m+1)}{n+2}$ is gotten from the graph in Fig 1.0,

then making n the subject of formula, we get the values of n

The number of plates falls off rapidly at first, though more slowly later, and a value of $R \approx 2$ is probably the most economic.

2. Data given



Total material Balance

$$F = D + B$$

$$100 = D + B$$

Component balance on octane

$$x_{F0}F = x_{D0}D + x_{B0}B$$

$$100 \times 0.6 = 0.98D + 0.125B$$

$$60 = 0.98D + 0.125B$$

$$\text{Recall: } 100 = D + B; \quad D = 100 - B$$

$$60 = 0.98(100 - B) + 0.125B$$

$$\therefore B = 44.44 \text{ kmol}$$

$$D = 100 - B = 100 - 44.44$$

$$D = 55.56 \text{ kmol}$$

The compositions and amounts of the streams are.

Component	Feed		Distillate		Bottoms	
	kmol	mol%	kmol	mol%	kmol	mol%
Ortho	60	60	54.44	98.0	5.56	12.5
Meta	4	4	0.33	0.6	3.67	8.3
Para	36	36	0.79	1.4	35.21	79.2
	100	100	55.56	100	44.44	100

Equations of operating lines

The liquid and vapour streams in the columns are obtained as follows.

Above the feed point: Liquid downflow, $L_n = 5D = 277.8$

Vapour up, $V_n = 6D = 333.4$

Below the feed point, assuming the feed is liquid at its boiling point.

Liquid downflow, $L_m = L_n + F = 277.8 + 100 = 377.8$

Vapour up, $V_m = L_m - W = 377.8 - 44.44 = 333.4$

The equations for the operating lines may then be written as:

Below the feed plate:

$$y_m = \frac{L_m}{V_m} x_{m+1} - \frac{W}{V_m} x_n$$

$$\text{ortho: } y_{mo} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.125)$$

$$y_{mo} = 1.133 x_{m+1} - 0.0166$$

$$\text{meta: } y_{mm} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.083)$$

$$y_{mm} = 1.133 x_{m+1} - 0.011$$

$$\text{Para: } y_{mp} = \left(\frac{377.8}{333.4} \right) x_{m+1} - \left(\frac{44.44}{333.4} \right) (0.792)$$

$$y_{mp} = 1.133 x_{m+1} - 0.105$$

Above the feed plate:

$$y_n = \frac{L_n}{V_n} x_{n+1} + \frac{D}{V_n} x_d$$

$$\text{ortho: } y_{no} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.98)$$

$$y_{no} = 0.833 x_{n+1} + 0.163$$

$$\text{meta: } y_{nm} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.006)$$

$$y_{nm} = 0.833 x_{n+1} + 0.001$$

$$\text{Para: } y_{np} = \left(\frac{277.8}{333.4} \right) x_{n+1} + \left(\frac{55.56}{333.4} \right) (0.014)$$

$$y_{np} = 0.833 x_{n+1} + 0.002$$

Composition of liquid on first plate

$$y_{s0} = \alpha_0 x_{s0} / \sum \alpha x_s$$

$$0.191 = 1.133 x_1 - 0.0166$$

$$x_1 = 0.183$$

Using iteration method, we have 16 ideal plates

3) Equilibrium data:

$$\text{kg A / kg B} \quad 0.05 \quad 0.10 \quad 0.15$$

$$\text{kg A / kg B} \quad 0.069 \quad 0.159 \quad 0.258$$

$$\text{Density (kg/m}^3\text{)} \quad S_A = 1200 \quad S_B = 1000 \quad S_S = 800$$

a) Countercurrent operation

Considering the solvent S, $160 \text{ cm}^3/\text{s} = 1.6 \times 10^{-4} \text{ m}^3/\text{s}$

and mass flow rate = $(1.6 \times 10^{-4} \times 800) = 0.128 \text{ kg/s}$

Solvent, $400 \text{ cm}^3/\text{s} = 4 \times 10^{-4} \text{ m}^3/\text{s}$

containing $a \text{ m}^3/\text{s}$ A and $(5 \times 10^{-4} - a) \text{ m}^3/\text{s}$ B

Thus: mass flow rate of A = $1200a \text{ kg/s}$

mass flow rate of B = $(4 \times 10^{-4} - a)1000 = (0.4 - 1000a) \text{ kg/s}$

Total = $(0.4 + 200a) \text{ kg/s}$

The concentration of the solution is

$$0.10 = \frac{1200a}{0.4 + 200a}$$

$$\therefore a = 3.39 \times 10^{-5} \text{ m}^3/\text{s}$$

4 Ca) For the pilot scale vessel:

$$C = \left(2.5 \times \frac{75}{100}\right) = 1.875 \text{ kg/m}^3$$

$$C_s = 2.5 \text{ kg/m}^3 \quad V = 1.0 \text{ m}^3 \quad t = 10 \text{ s}$$

$$C = C_s (1 - e^{-(K'A/V)t})$$

$$1.875 = 2.5 (1 - e^{-(\frac{K'A}{100})10})$$

$$\frac{K'A}{b} = 0.139 \text{ m}^3/\text{s}$$

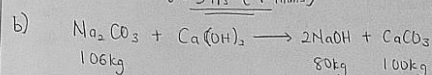
For the full scale vessel

$$C = \left(500 \times \frac{28}{100}\right) = 1.40 \text{ kg/m}^3$$

$$C_s = 2.5 \text{ kg/m}^3 \quad V = 100 \text{ m}^3$$

$$1.40 = 2.5 (1 - e^{-\frac{0.139 b t}{100}})$$

$$t = 591 \text{ s} \quad (9.9 \text{ mins})$$



If x'_1 , x'_2 and x'_3 are the solute: solvent ratios in thickness 1, 2, and 3 respectively, the quantities of CaCO_3 , NaOH and water in each of the streams can be calculated for every 100 kg of calcium carbonate.

	CaCO ₃	NaOH	Water
Overall balance			
Feed from reactor	100	80	900
Feed as wash water	—	—	W _f
Product underflow	100	300x' ₃	300
Product overflow	—	80 - 300x' ₃	600 + W _f
Thickener 1			
Feed from reactor	100	80	900
Feed overflow	—	300(x' ₁ - x' ₂)	W _f
Product underflow	100	300x' ₁	300
Product overflow	—	80 - 300x' ₁	600 + W _f
Thickener 2			
Feed underflow	100	300x' ₁	300
Feed overflow	—	300(x' ₁ - x' ₂)	W _f
Product underflow	100	300x' ₂	300
Product overflow	—	300(x' ₁ - x' ₂)	W _f
Thickener 3			
Feed underflow	100	300x' ₂	300
Feed water	—	—	W _f
Product underflow	100	300x' ₃	300
Product overflow	—	300(x' ₂ - x' ₃)	W _f

Since the final underflow must contain only 1 percent of NaOH
 then: $\frac{300x'_3}{100} = 0.01$

If the equilibrium is achieved in each of the thickeners, the ratio of NaOH to water will be the same in the underflow and the overflow and:

$$\frac{300(x'_2 - x'_3)}{W_f} = x'_3$$

$$\frac{300(x'_1 - x'_2)}{W_f} = x'_2$$

$$\frac{80 - 300x'_3}{600 + W_f} = x'_1$$

Solution of these four simultaneous equations gives:

$$x'_3 = 0.0033 \quad x'_2 = 0.0142 \quad x'_1 = 0.05 \quad W_f = 980$$

The amount of water required for washing 100 kg CaCO₃ is 980 kg. The solution fed to reactor contains 0.25 kg / kg CaCO₃. This is equivalent to 0.236 kg / kg CaCO₃ and hence the actual water required.

$$= \frac{980 \times 0.236}{100} = 0.23 \text{ kg / kg}$$

5 (a) With a single feed and bleed stage.

Where: Q₀ = Volumetric flowrate of feed
 Q₂ = Volumetric flowrate of concentrate
 C₀ = Solute concentration in the feed

C_2 = Solute concentration in the concentrate, F
 A = required membrane area
 NB: It is assumed that there is no loss of solute through the membrane.
 The concentration (C_1) at which the flux becomes fouling-limited is

$$0.04 = 0.02 \ln \left(\frac{30}{C_1} \right)$$

$$C_1 = 4 \text{ kg/m}^3$$
 That is, below this concentration the membrane flux is 0.04 m/h
 This does not pose a constraint for the single stage as the concentration of solute C_2 will be that of the final concentration, 20 kg/m^3
 Conservation of solute gives: $Q_0 C_0 = Q_2 C_2$
 Fluid balance: $Q_0 = F + Q_2$
 Combining the equations and substituting known values

$$2.438 = A \cdot 0.02 \ln \left(\frac{30}{20} \right)$$

$$A = 301 \text{ m}^2$$
 Thus 10 modules will almost meet the specification for the single stage.

b) With two feed and bleed stages in series

 Q_1 = Volumetric flowrate of retentate at the intermediate point
 C_1 = Concentration of solute in the retentate
 A_1 and A_2 are the area of membranes

$\therefore$ Mass flowrate of $A = 1200 \times 3.39 \times 10^{-5}$
 $= 0.041 \text{ kg/s}$
 mass flowrate of $B = (0.4 - 1000 \times 3.39 \times 10^{-5})$
 $= 0.366 \text{ kg/s}$
 Ratio of A to B in the feed $= \frac{0.041}{0.366} = 0.112 \text{ kg/kg}$
 The ratio, $X_f = 0.112 \text{ kg/kg}$ is marked in.
 $\therefore$ Mass flowrate of $B = \frac{0.366}{0.128} = 2.86$
 mass flowrate of S

Since pure solvent is added $Y_{n+1} = Y_n = 0$ and a line of slope 2.86 is drawn in such that off from $X_f = 0.112 \text{ kg/kg}$ to $Y_4 = 0$ gives exactly three stages.
 When $Y_4 = 0$, $X_0 - X_3 = 0.057 \text{ kg/kg}$
 Hence, the composition of final raffinate $= 0.057 \text{ kg A / kg B}$

b) Multiple contact

 $\therefore \frac{0.128}{3} = 0.0427 \text{ kg/s}$ of pure solvent S is fed to each stage

Stage 1 $X_f = \frac{0.041}{0.366} = 0.112 \text{ kg/kg}$

From the equilibrium curve, the extract contains 0.18 A/kg S and
 $0.18 \times 0.0427 = 0.0077 \text{ kg/s A}$

Raffinate from stage 1 contains: $0.041 - 0.0077 = 0.0333 \text{ kg/s A}$
 0.366 kg/s B

$$X_1 = \frac{(0.0333)}{0.366} = 0.091 \text{ kg/kg}$$

Stage 2

$$X_1 = 0.091 \text{ kg/kg}$$

From graph, this contains 0.14 kg A/kg S
 $\therefore 0.14 \times 0.0427 = 0.006 \text{ kg/s A}$

The raffinate from stage 2 contains: $0.0333 - 0.006 = 0.0273 \text{ kg/s A}$
 and 0.366 kg/s B

$$X_2 = \frac{0.0273}{0.366} = 0.075 \text{ kg/kg}$$

Stage 3

$$X_2 = 0.075 \text{ kg/kg}$$

The extract contains 0.114 kg A/kg S
 $0.114 \times 0.0427 = 0.0049 \text{ kg/s A}$

The raffinate from stage 3 contains: $0.0273 - 0.0049 = 0.0224 \text{ kg/s A}$
 and 0.366 kg/s B

$$X_3 = \frac{0.0224}{0.366} = 0.061 \text{ kg/kg} \text{ (final composition)}$$

Conservation of solute: $Q_0 C_0 = Q_1 C_1 = Q_2 C_2$

A fluid balance on stage 1: $Q_0 = Q_1 + F_1$

A fluid balance on stage 2: $Q_1 = Q_2 + F_2$

Substituting given values in equations above

$$2.5 = \frac{1.25}{C_1} + 0.02 \text{ A, } \ln\left(\frac{30}{C_1}\right)$$

$$\frac{1.25}{C_1} = 0.0625 + 0.00811 \text{ A}_1$$

The procedure is to use trial by error to eliminate the value of C_1 that gives the optimum values of A_1 and A_2 . Thus;

If $C_1 = 5 \text{ kg/m}^3$, then $A_1 = 63 \text{ m}^2$ and $A_2 = 23 \text{ m}^2$

An arrangement of 3 modules - 1 module is required

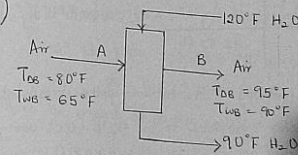
If $C_1 = 4 \text{ kg/m}^3$, then $A_1 = 55 \text{ m}^2$ and $A_2 = 31 \text{ m}^2$

An arrangement of 2 modules - 1 module is almost sufficient

If $C_1 = 4.5 \text{ kg/m}^3$, then $A_1 = 59 \text{ m}^2$ and $A_2 = 27 \text{ m}^2$

An arrangement of 2 modules - 1 module which meets the requirement

6)



A = inlet air, B = exit air

T_{0S} = Dry bulb temperature

T_{0B} = Wet bulb temperature

From the psychrometric chart, we can determine the amount of water in dry air, enthalpy, humidity, and humid volume for both A and B, which are given as

	A	B
Water in dry air = $\frac{H(\text{lb H}_2\text{O})}{\text{lb dry air}}$	0.0098	0.0297

Enthalpy = $\Delta H \left(\frac{\text{Btu}}{\text{lb dry air}} \right)$	29.93	55.83
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Humidity = $\left(\frac{\text{grains H}_2\text{O}}{\text{lb dry air}} \right)$	69	208
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Humid volume = $\hat{V} \left(\frac{\text{ft}^3}{\text{lb dry air}} \right)$	13.82	14.65
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In calculating for the cooling water exit rate, we have to determine an energy balance around the process:

Given the basis: $8.30 \times 10^5 \text{ ft}^3/\text{hr}$ of moist air

Converting the basis to lb/hr

$$\frac{8.30 \times 10^5 \text{ ft}^3}{\text{hr}} \times \frac{\text{lb dry air}}{13.82 \text{ ft}^3} = 6.00 \times 10^5 \text{ lb dry air/hr}$$

Finding the relative enthalpy of the entering water stream, taking the reference temperature of water to be 32°F (0°C), using the enthalpy equation which is:

$$\Delta H = C_{p_{H_2O}} \Delta T, \quad C_{p_{H_2O}} = 1$$

$$\Delta H = 1(120 - 32) = 88 \text{ Btu/lb H}_2\text{O}$$

The relative enthalpy of the exit stream

$$\Delta H = C_{p_{H_2O}} \Delta T = 1(90 - 32) = 58 \text{ Btu/lb H}_2\text{O}$$

Taking the loss of water through the cooling tower to be

$$0.0297 - 0.0098 = 0.0199 \text{ lb H}_2\text{O/lb dry air}$$

Taking the material balance of the water stream to be

$$W - 0.0199 = \text{lb H}_2\text{O leaving tower in the water stream}$$

lb dry air

Taking the energy/enthalpy balance around the entire process to be

air and water in air entering + water stream entering =
air and water in air leaving + water stream leaving.

$$\frac{29.93 \text{ Btu}}{\text{lb dry air}} \times \frac{6.00 \times 10^5 \text{ lb dry air}}{\text{hr}} + \frac{88 \text{ Btu}}{\text{lb H}_2\text{O}} \times \frac{W \text{ lb H}_2\text{O}}{\text{lb dry air}} \times \frac{6 \times 10^5 \text{ lb dry air}}{\text{hr}}$$

$$= \frac{55.83 \text{ Btu}}{\text{lb dry air}} \times \frac{6 \times 10^5 \text{ lb dry air}}{\text{hr}} + \frac{58 \text{ Btu}}{\text{lb H}_2\text{O}} \times \frac{(W - 0.0199) \text{ lb H}_2\text{O}}{\text{lb dry air}} \times \frac{6 \times 10^5 \text{ lb dry air}}{\text{hr}}$$

$$(29.93 + 88W)(\text{Btu})(6 \times 10^5) = 55.83 + 58(W - 0.0199)(6 \times 10^5)(\text{Btu})$$

$$29.93 + 88W = 55.83 + 58(W - 0.0199)$$

$$W = 0.825 \text{ lb H}_2\text{O/lb dry air}$$

There the amount of water cooled in lb/hr will be given as:

$$\frac{0.825 \text{ lb H}_2\text{O}}{\text{lb dry air}} \times \frac{6 \times 10^5 \text{ lb dry air}}{\text{hr}} = 4.95 \times 10^5 \text{ lb H}_2\text{O/hr}$$

7) When pellet diffusion is dominant, assuming $C_m = 0$ and the fluid concentration outside the pellet is constant, then

$$\frac{C_s}{C_s^*} = 1 - \frac{6}{\pi^2} \exp\left[-(\Delta_e \pi^2 t / r_i^2)\right]$$

Hence a plot of $\ln\left[1 - \frac{C_s}{C_s^*}\right]$ against t should be linear.

For $r = r_i$ with $C_s^* = 0.132 \text{ kg/kg}$ then

$t \text{ (min)}$	$C_s \text{ (kg/kg)}$	C_s/C_s^*	$1 - (C_s/C_s^*)$
2	0.091	0.69	0.31
4	0.097	0.73	0.27
10	0.105	0.80	0.20
20	0.113	0.86	0.14
40	0.125	0.95	0.05
60	0.128	0.97	0.03
120	0.132	1.0	0

The data are plotted in a graph which confirms the linearity and from which $\frac{\pi^2 \Delta_e}{r_i^2} = 0.043$

For a pellet of twice the radius, that is $r = 2r_i$

$$\text{and the slope} = (-0.043/4) = -0.011$$

Thus, when the radius = $2r_i$

$$\frac{C_s}{C_s^*} = 1 - \frac{6}{\pi^2} \exp(-0.011t)$$

Alternatively, we may be made of the quadratic driving force equation

$$\frac{dC_s}{dt} = \frac{k \Delta_e}{r_i^2} \frac{C_s^{*2} - C_s^2}{2(C_s - C_{s0})} \quad \text{--- (1)}$$

Integrating from the initial condition, $t = 0$ and $C_s = 0$, to t , then

$$\frac{C_s}{C_s^*} = \left[1 - \exp\left(-\frac{k \Delta_e t}{r_i^2}\right)\right]^{0.5}$$

Indicating that a plot of $\ln\left[1 - \left(\frac{C_s}{C_s^*}\right)^2\right]$ against t should also be linear.

$t \text{ (min)}$	$C_s \text{ (kg/kg)}$	(C_s/C_s^*)	$1 - \left(\frac{C_s}{C_s^*}\right)^2$
2	0.091	0.69	0.52
4	0.097	0.73	0.47
10	0.105	0.80	0.36
20	0.113	0.86	0.26
40	0.125	0.95	0.10
60	0.128	0.97	0.06
120	0.132	1.0	0

$$\text{Therefore: } \frac{k \Delta_e}{r_i^2} = 0.04$$

$$\text{For a pellet twice the size } \frac{C_s}{C_s^*} = \left[1 - \exp(-0.01t)\right]^{0.5} \quad \text{--- (1)}$$

Values of C_s for radius = $2r_i$ are calculated from equ (1) and (1) to give the following results

$t \text{ (min)}$	$C_s \text{ (kg/kg)}$	
	equ (1)	equ (1)
4	0.055	0.026
20	0.068	0.056
60	0.091	0.088

